

Joint inversion

Multigeophysical inversion

Ketil Hokstad, November 2018

Ketil Hokstad, April 2021

Ketil Hokstad, March 2025

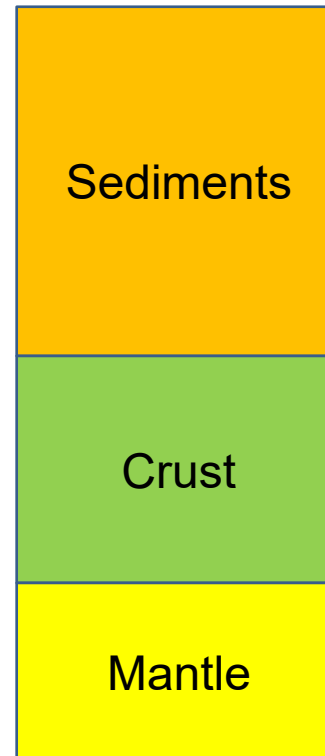
Joint inversion and multigeophysical inversion

Two main types of objectives

- Structural imaging
- Estimating properties of interest

Many scales and applications

- Petroleum exploration
 - Structural imaging
 - Fluid prediction
 - Basin modeling
- Geothermal exploration
- Mineral exploration
- Academic applications



- AVO and CSEM inversion
 - Prospect risking
- PSDM: Tomography/FWI, MT, FTG inversion
 - Improved imaging
- Crustal properties: Gravity, magnetic, MT, refractions
 - COB, RHP, heat flow
 - Hydrothermal systems
- Mantle properties: Gravity, topography, geoid
 - Depth to LAB => heat flow

From joint interpretation to joint inversion

We have always done joint interpretation in exploration

Petroleum

«Petroleum system analysis»

- Source
- Reservoir
- Trap/Seal



Geothermal

«Play fairway analysis»

- Heat source
- Recharge
- Producibility



Seabed massive sulfides

«Mineral systems analysis»

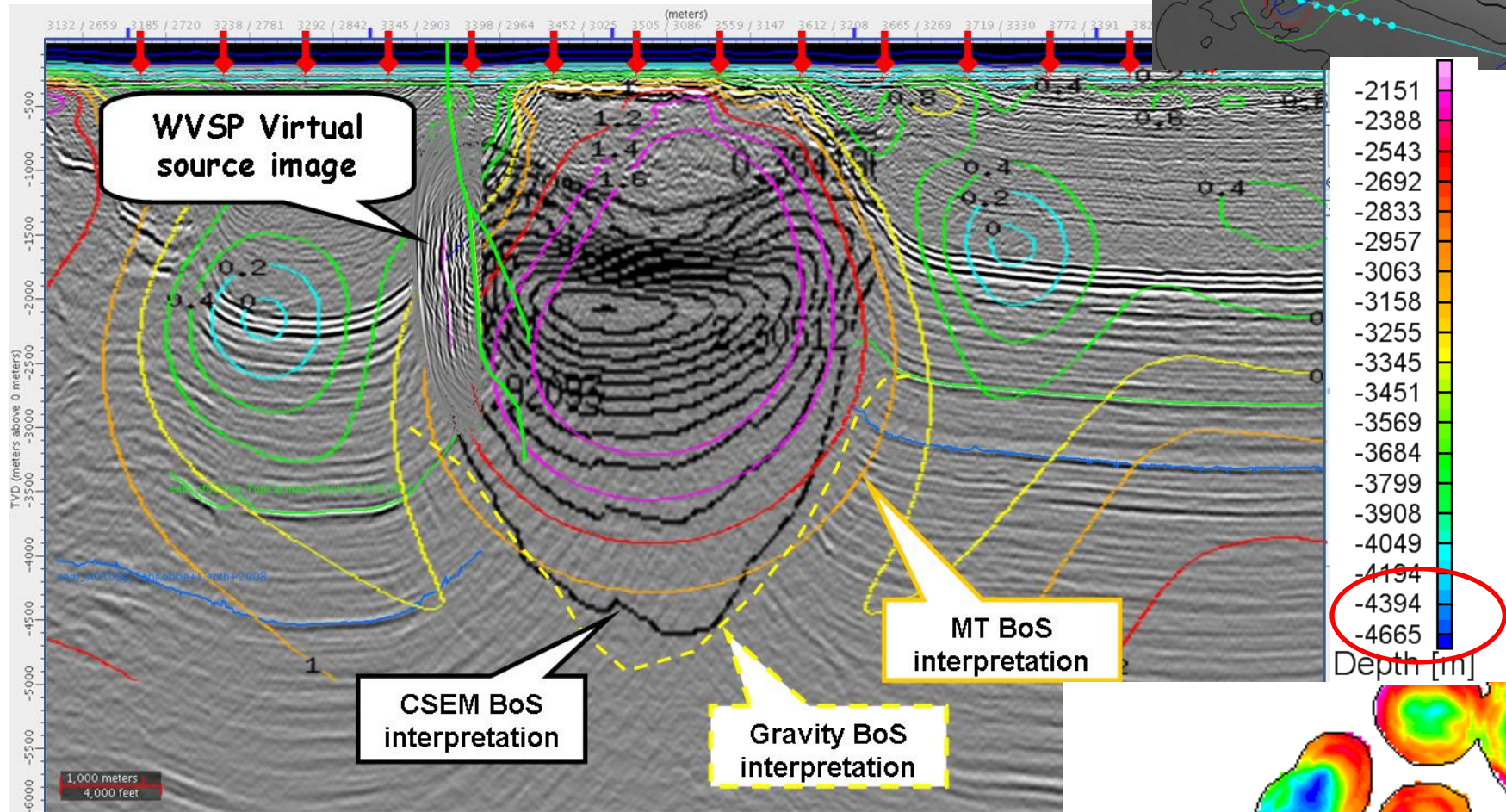
- Metal source
- Fluid pathway
- Accumulation and preservation



If one play element fails, the play fails

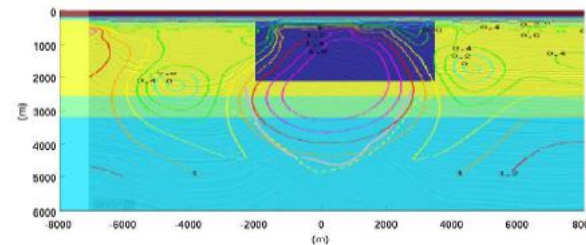
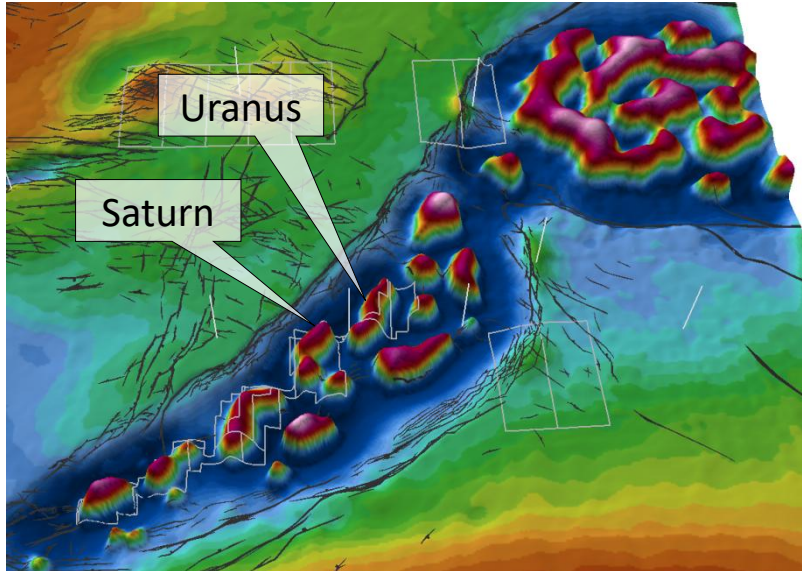
From joint interpretation to joint inversion

Uranus well line – joint interpretation

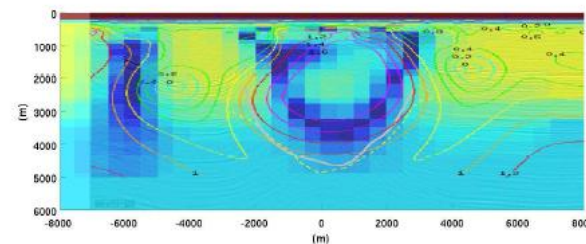


Hokstad et al. (2011), Joint Imaging of Geophysical Data:
Case History from the Nordkapp Basin, Barents Sea

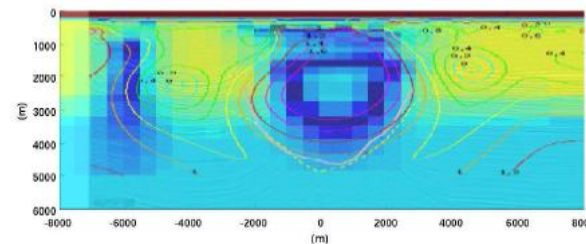
Contrast source inversion (Wiik et al, 2013): Nordkapp Basin joint CSEM and MT inversion



Initial model



MT inversion
(E and H fields)



MT+CSEM inversion

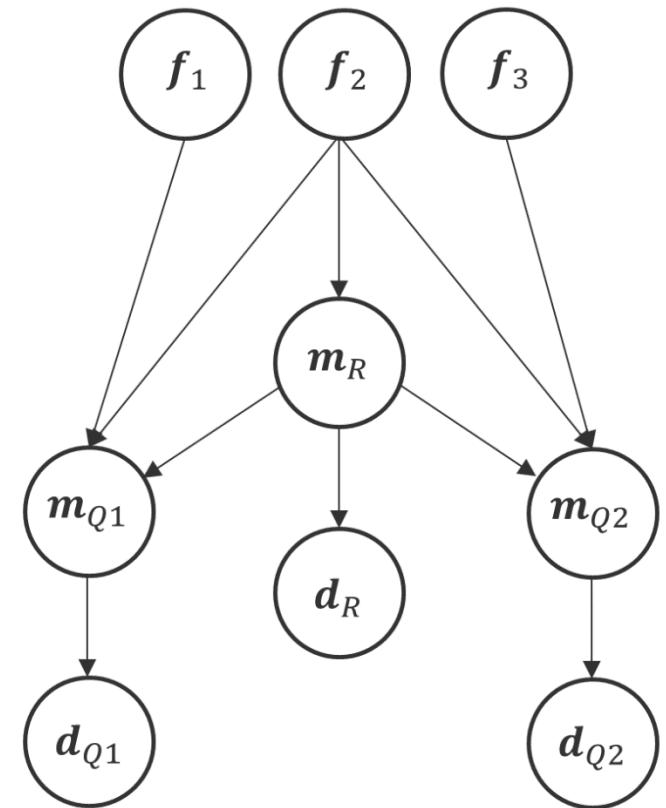
Dominating contributions;

- CSEM => vertical resistivity
- MT => horizontal resistivity

Generic SJI/MGI formalism

Two different approaches

- Statistical
 - Bayesian inversion
 - Including posterior uncertainty
- Optimization
 - BFGS, L-BFGS
 - Gauss Newton ...
- Both can be derived from a common starting point
- **Generic formulation for any multi-geophysical inverse problem**



Bayesian network (DAG) where model m_R acts as a structural reference for models m_{Qi}

Simultaneous Joint Inversion (SJI)

Cross-gradient principle



Orthogonal vectors:

$$\mathbf{a} \cdot \mathbf{b} = 0$$

$$a_i b_i = 0$$



Parallel vectors:

$$\mathbf{a} \times \mathbf{b} = (0,0,0)$$

$$\epsilon_{ijk} a_j b_k = 0$$

Cartesian model gradient

$$\mathbf{h}_Q = \nabla \mathbf{m}_Q(\mathbf{x})$$

Cross-gradient objective function

$$\Phi_{XG} = [\mathbf{h}_R \times \mathbf{h}_Q]^T \boldsymbol{\Sigma}_{XG}^{-1} [\mathbf{h}_R \times \mathbf{h}_Q]$$

Promote structural similarity of different geophysical models

$$p(\mathbf{m}_Q | \mathbf{m}_R) \propto e^{-\Phi_{XG}}$$

X-gradient
objective function

SJI objective function – structural constraints

$$\Phi_{SJI} = \Phi_{MLH} + \Phi_{pri} + \Phi_{sm} + \Phi_{RP} + \Phi_{XG}$$

Single-domain inversions:

- Data fit (MLH)
- Model prior
- Smoothness

SJI terms:

- Rock physics models (Gardner, mudrock)
- Cross-gradients

$$\Phi_{XG} = \frac{1}{2} \sum_{Q \neq R} [\nabla \mathbf{m}_R \times \nabla \mathbf{m}_Q]^T \Sigma_{XGQ}^{-1} [\nabla \mathbf{m}_R \times \nabla \mathbf{m}_Q] \quad \text{X-gradient}$$

$$\Phi_{RP} = \frac{1}{2} \sum_Q [\mathbf{m}_Q - \mathcal{G}_Q(\mathbf{m}_R)]^T \Sigma_{RPQ}^{-1} [\mathbf{m}_Q - \mathcal{G}_Q(\mathbf{m}_R)] \quad \text{Rock physics}$$

Optimization approach – least squares

- Minimization of total SJI objective function Φ_{SJI}

$$\Delta \mathbf{m} = -\mathbf{H}^{-1} \mathbf{g}$$

$$\mathbf{g} = \frac{\partial \Phi}{\partial \mathbf{m}} \quad \text{gradient}$$

$$\mathbf{H} = \frac{\partial^2 \Phi}{\partial \mathbf{m}^2} \quad \text{Hessian}$$

- Written out:

$$\Delta \mathbf{m}_Q = -[\mathbf{H}_Q^{MLH} + \mathbf{H}_Q^{pri} + \mathbf{H}_Q^{sm} + \mathbf{H}_Q^{RP} + \mathbf{H}_Q^{XG}]^{-1} [\mathbf{g}_Q^{MLH} + \mathbf{g}_Q^{pri} + \mathbf{g}_Q^{sm} + \mathbf{g}_Q^{RP} + \mathbf{g}_Q^{XG}]$$

- Need to broadcast gradients and Hessians from single-domain inversions
- Finds the (local) minimum of the total SJI objective function

Simplified SJI approach

- Simplified split-step approach:

1. Single domain inversion

$$\Delta \tilde{\mathbf{m}}_Q = -[\mathbf{H}_Q^{MLH} + \mathbf{H}_Q^{pri} + \mathbf{H}_Q^{sm}]^{-1}[\mathbf{g}_Q^{MLH} + \mathbf{g}_Q^{pri} + \mathbf{g}_Q^{sm}]$$

2. SJI update

$$\Delta \mathbf{m}_Q = \Delta \tilde{\mathbf{m}}_Q - [\mathbf{H}_Q^{RP} + \mathbf{H}_Q^{XG}]^{-1}[\mathbf{g}_Q^{RP} + \mathbf{g}_Q^{XG}]$$

- Does not need to broadcast single domain-gradients and Hessians
- Does generally not find the (local) minimum of the full SJI objective function

X-gradient terms - details

Model-parameter gradient:

$$\mathbf{g}_Q^{XG} = [\mathbf{h}_R \times \mathbf{h}_Q]^T \Sigma_{XG}^{-1} [\mathbf{h}_R \times \frac{\partial \mathbf{h}_Q}{\partial \mathbf{m}_Q}]$$

Hessian (Gauss-Newton):

$$\mathbf{H}_Q^{XG} = \left[\mathbf{h}_R \times \frac{\partial \mathbf{h}_Q}{\partial \mathbf{m}_Q} \right]^T \Sigma_{XG}^{-1} \left[\mathbf{h}_R \times \frac{\partial \mathbf{h}_Q}{\partial \mathbf{m}_Q} \right]$$

Cartesian gradient and derivatives:

$$\mathbf{h}_Q = \nabla \mathbf{m}_Q$$

$$\frac{\partial h_{Qki}}{\partial m_{Qk}} = \sum_{j=1}^3 \frac{\partial h_{Qki}}{\partial x_j} \frac{\partial x_j}{\partial m_{Qk}} = \sum_{j=1}^3 \frac{1}{h_{Qkj}} \frac{\partial h_{Qki}}{\partial x_j}$$

$$Q \in \{\rho, v_P, v_S, R, M, \dots\}$$

$$k \in \{1, 2, \dots, N_{Qm}\}$$

$$i, j \in \{1, 2, 3\}$$

$$\frac{\partial h_{Qki}}{\partial m_{Qk}} = \frac{\partial}{\partial x_i} \ln \left(\prod_{j=1}^3 h_{Qkj} \right)$$

Published work on SJI

Moorkamp et al (GJI, 2011) and di Stefano et al (Geophysics, 2011) use a similar X-gradient objective function as us:

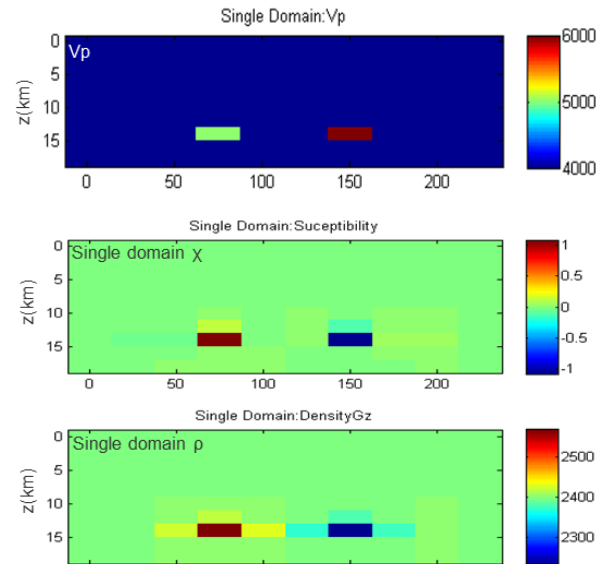
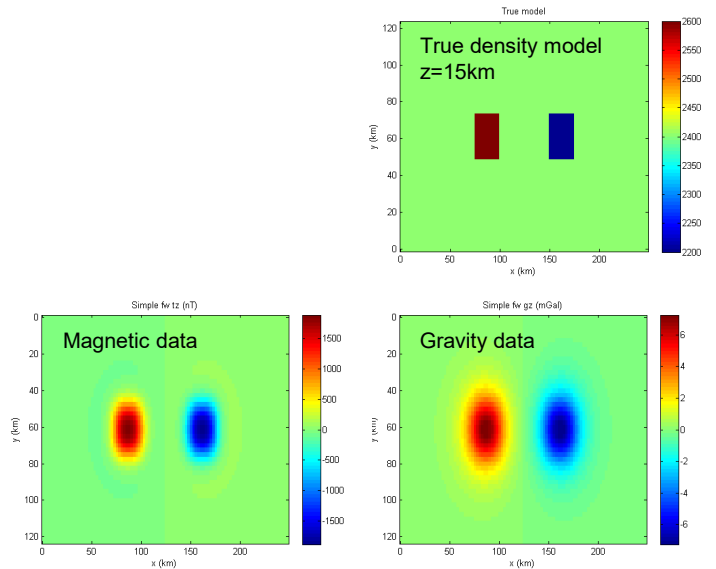
$$\Phi_{XG} = \frac{1}{2} \sum_{Q \neq R} [\nabla \mathbf{m}_R \times \nabla \mathbf{m}_Q]^T \boldsymbol{\Sigma}_{XGQ}^{-1} [\nabla \mathbf{m}_R \times \nabla \mathbf{m}_Q] \quad \text{X-gradient}$$

Gallardo (GRL, 2007) use a formulation based on Lagrange multipliers:

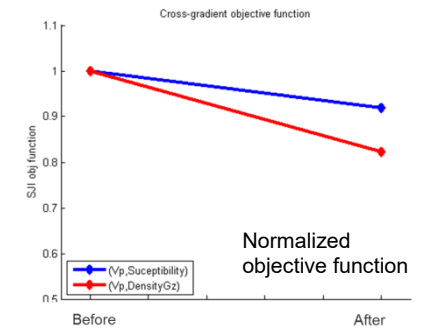
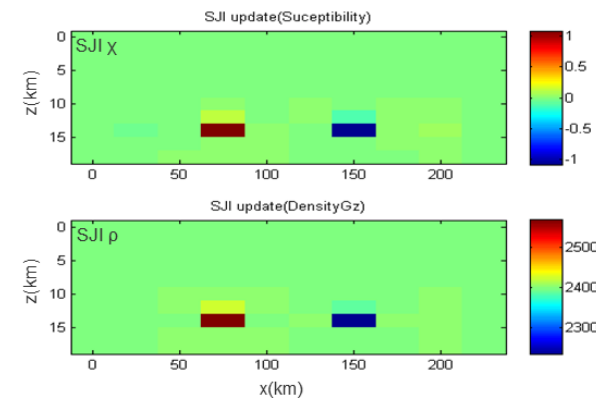
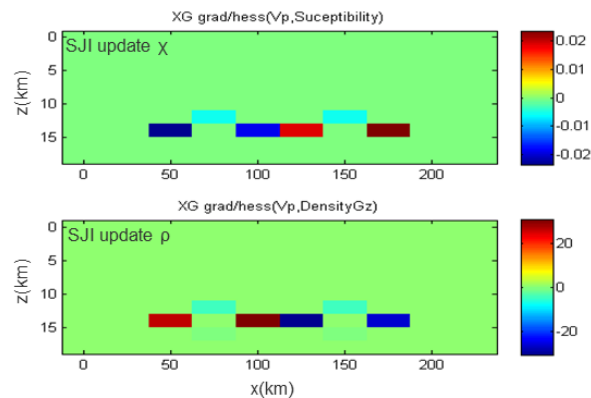
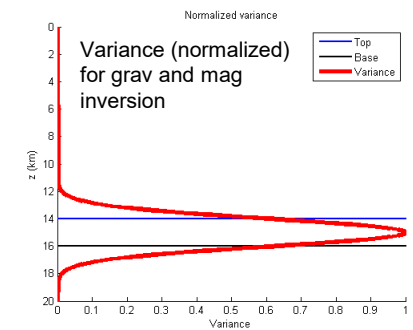
$$\min \left\{ \sum_{i=1} \phi_{di} + \sum_{j=1} \phi_{Lj} + \sum_{p=1} \phi_{0p} \right\} \quad \text{X-gradient}$$

subject to $\tau_g = 0$.

Synthetic example: Joint grav-mag inversion



Velocity model: Structural constraint (not updated in the joint inversion)



Nordkapp Basin example: Cross-gradient

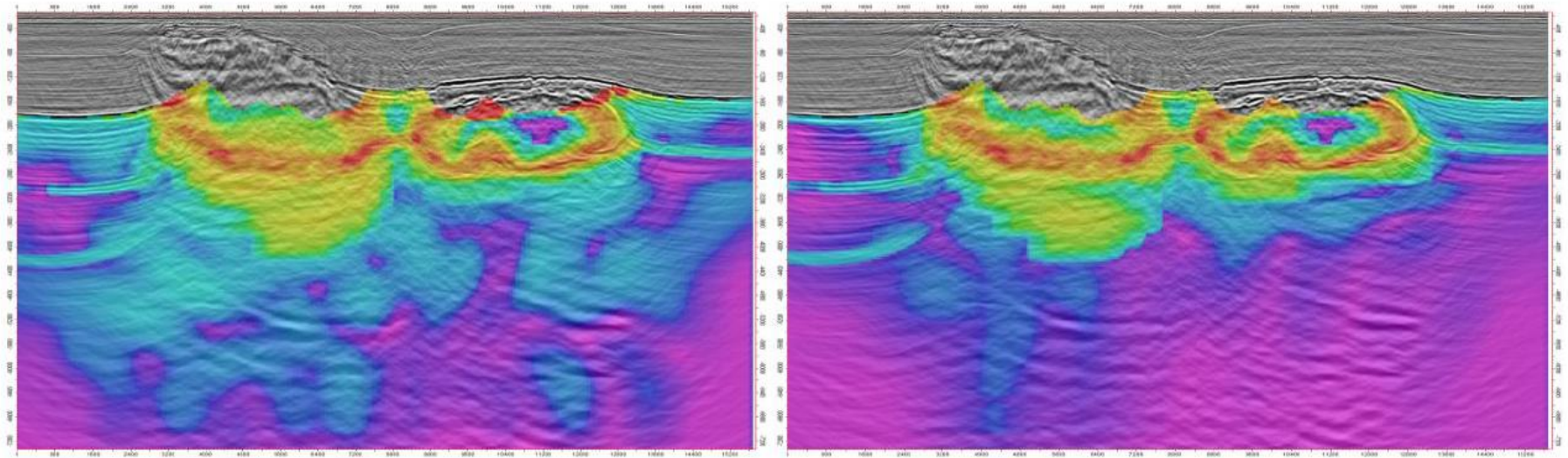


Figure 6 Cross-Gradient functional display between single domain seismic and gravity inversion results (left), and between SJI velocity and density property (right). The overall decrease of Cross-Gradient indicates a more adherent solution to structural concordance hypothesis for SJI.

Nordkapp Basin example: PSDM vs SJI

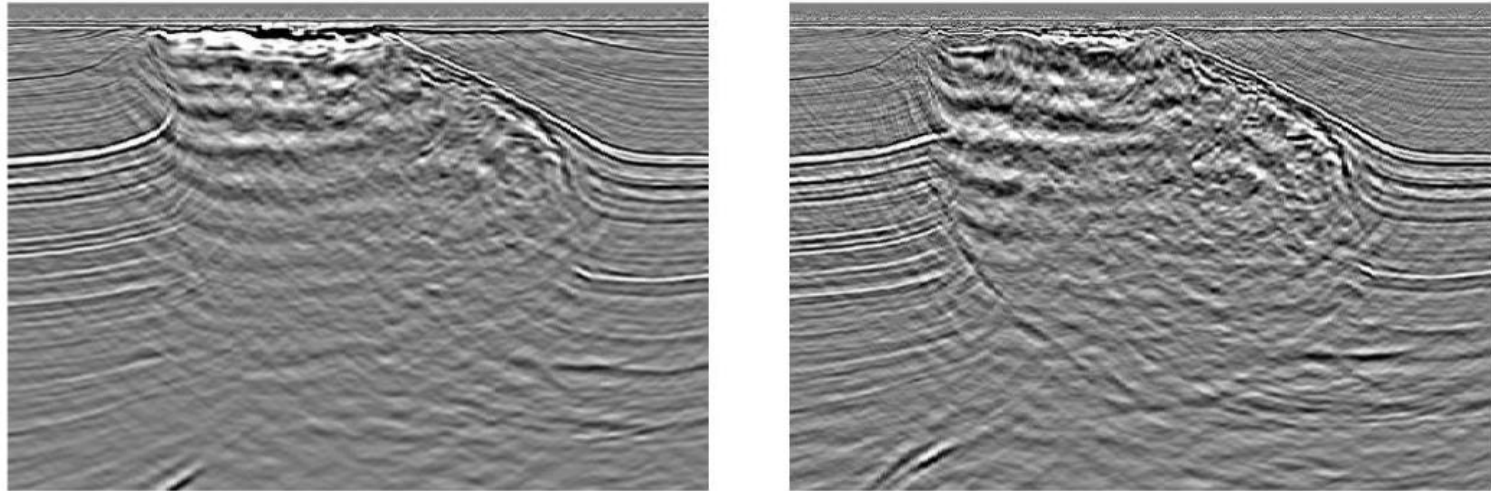
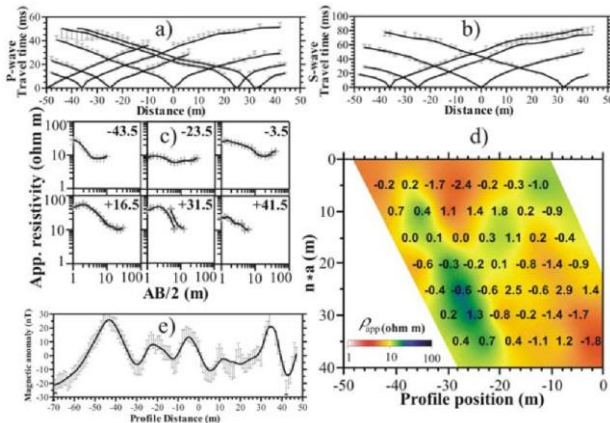


Figure 5 *Seismic tomography image before SJI (left) and SJI image after grid based Cross Gradient update (right).*

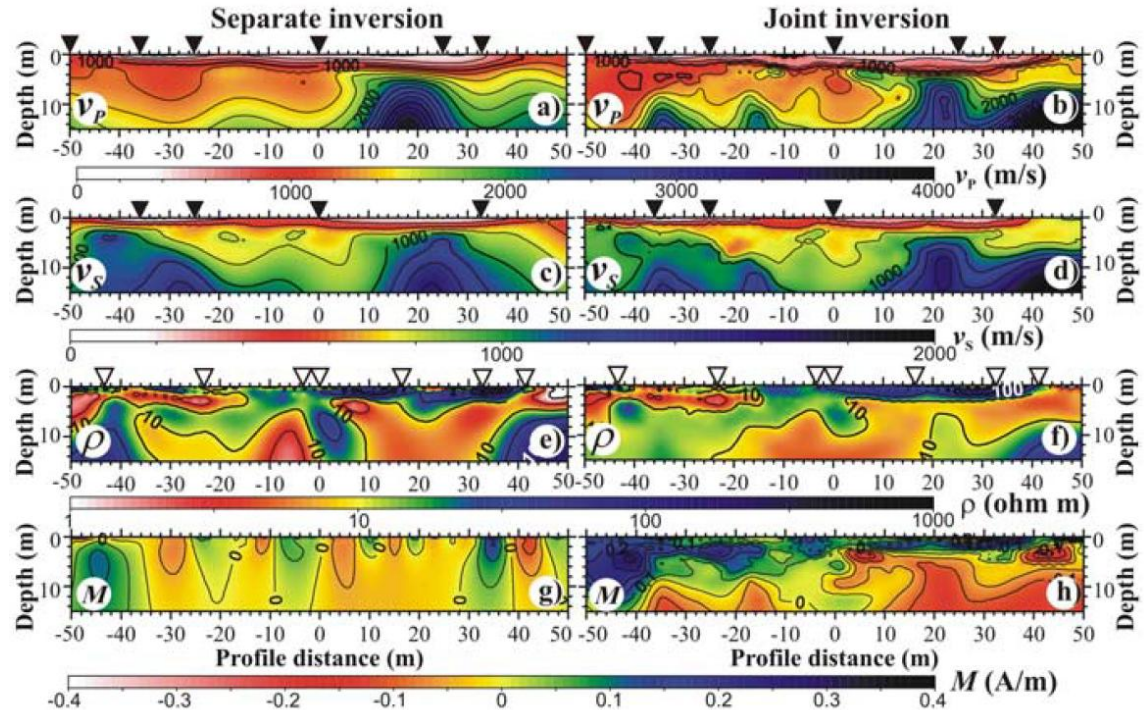
From Mantovani et al, EAGE (2014) and diStefano et al (2011)

SJI example from Gallardo (GRL, 2007)



Data:

- a) P-wave traveltimes
- b) S-wave traveltimes
- c) Vertical DC electric
- d) Dipole-dipole EM
- e) Magnetic data

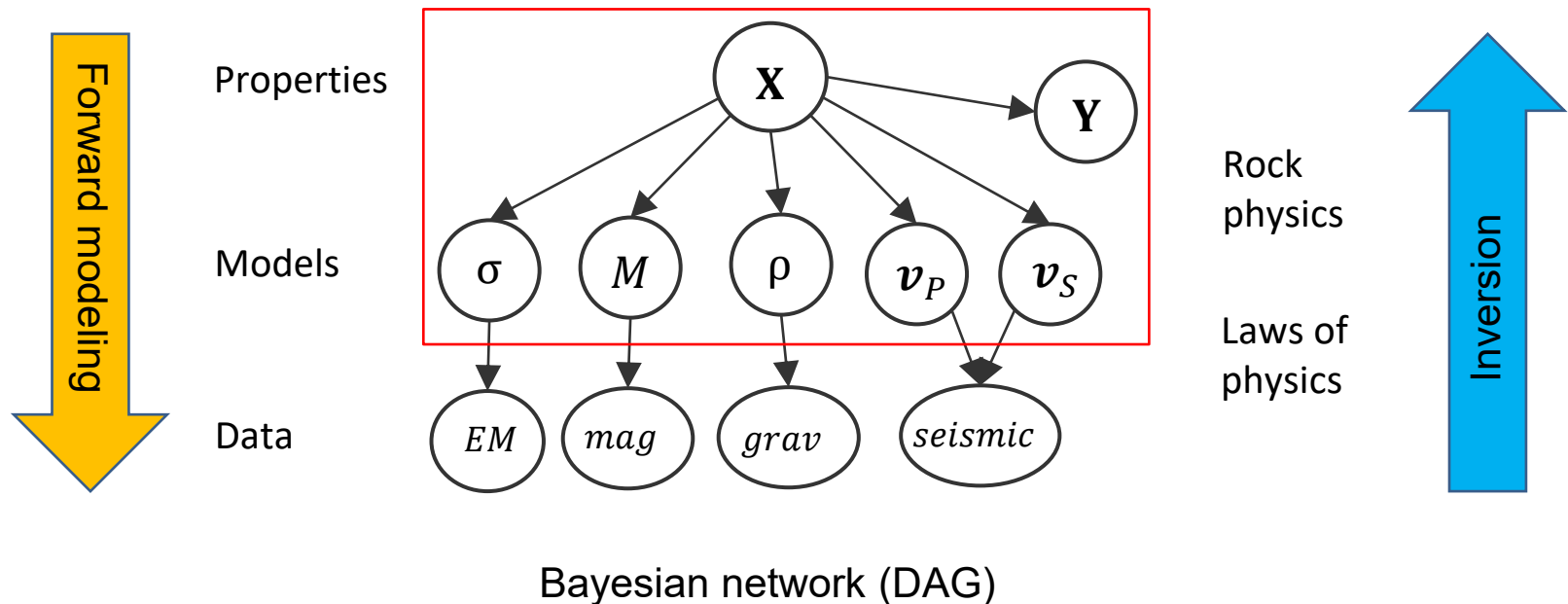


Single domain
inversions

Joint inversion

Multigeophysical inversion

Estimating properties of interest

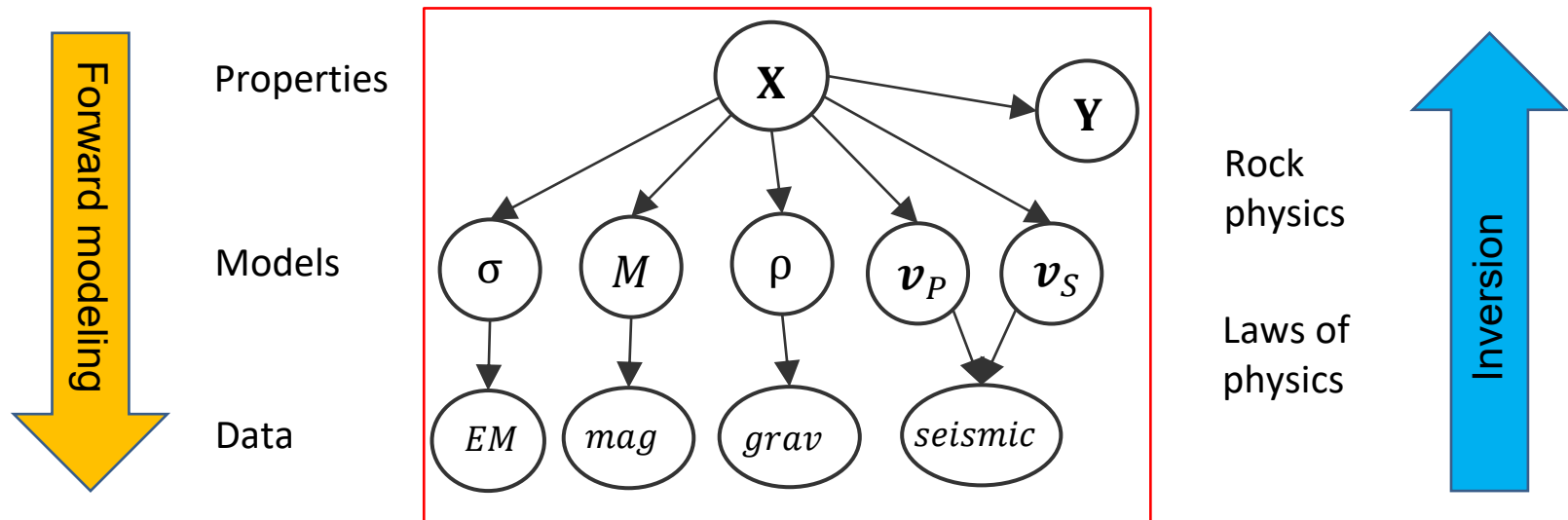


$$p(\mathbf{X}|m_1, \dots, m_n) = c \prod_{i=1}^n p(m_i|\mathbf{X})p(\mathbf{X}; \lambda)$$

- Conditional independence
- Independence of noise

Multigeophysical inversion

Estimating properties of interest

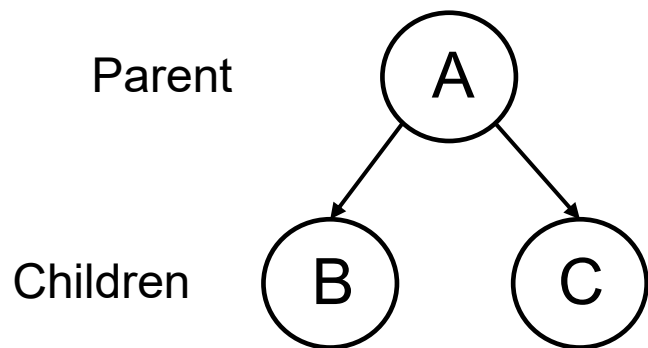


Complete expression; marginalization

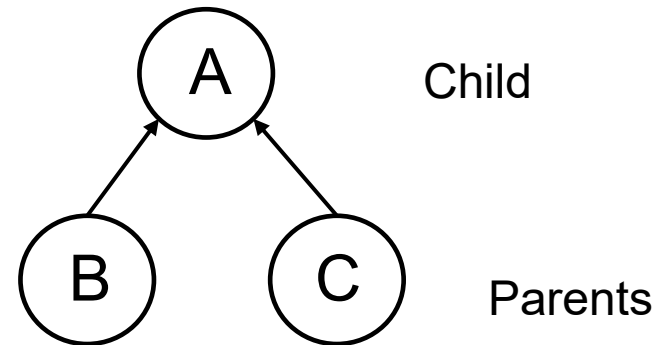
$$p(\mathbf{X}|\mathbf{d}) = \int p(\mathbf{X}|\mathbf{m})p(\mathbf{m}|\mathbf{d})d\mathbf{m}$$

Bayesian networks – the general formula

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i | x_i^{pa})$$



$$p(A, B, C) = p(B|A)p(C|A)P(A)$$

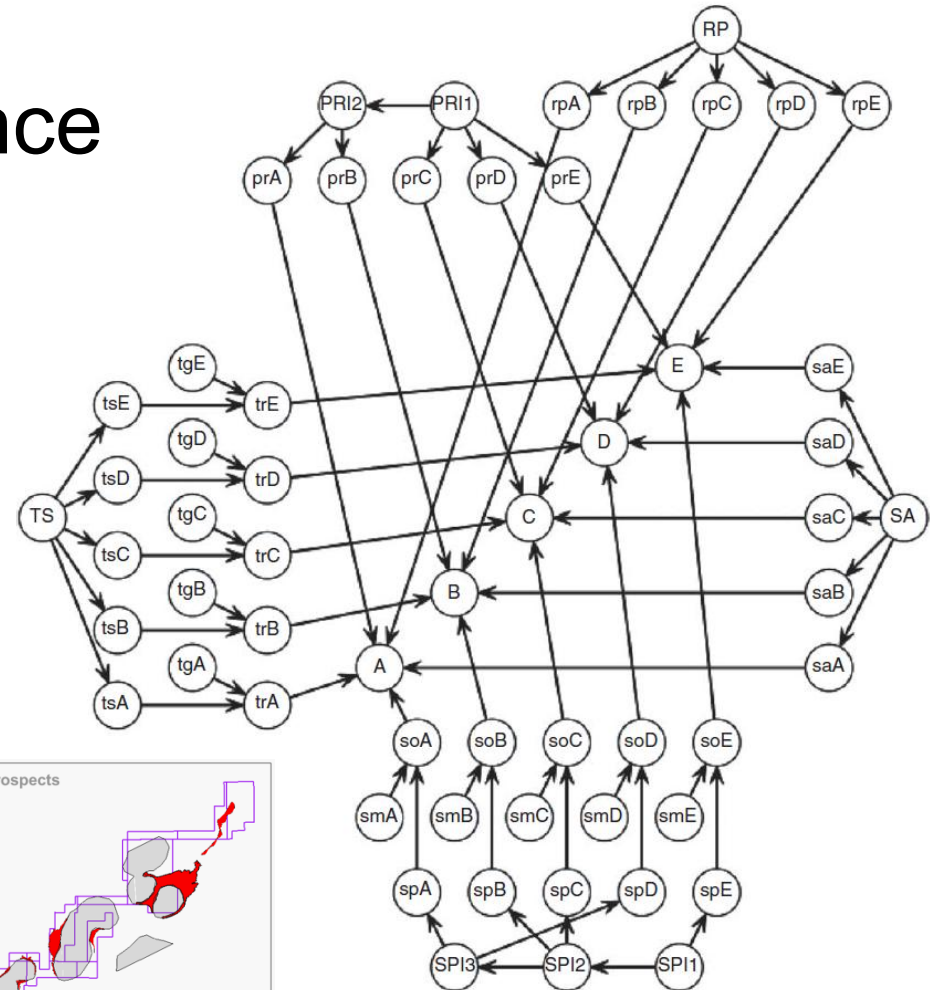
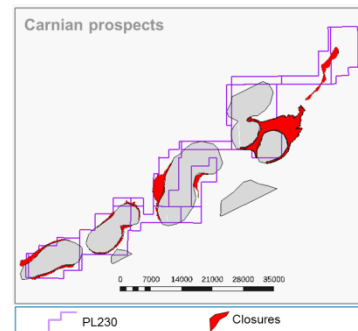
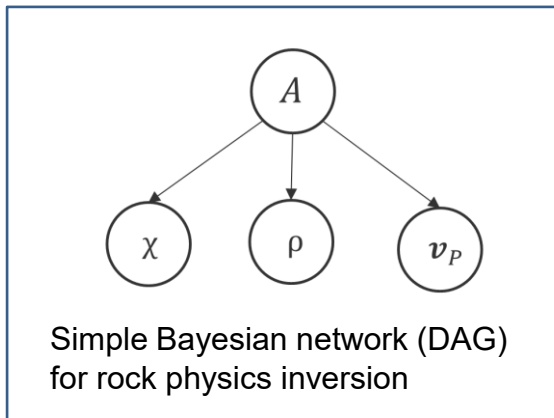


$$p(A, B, C) = p(A|B, C)P(B, C)$$

Bayesian networks; conditional independence

Joint probability factorization

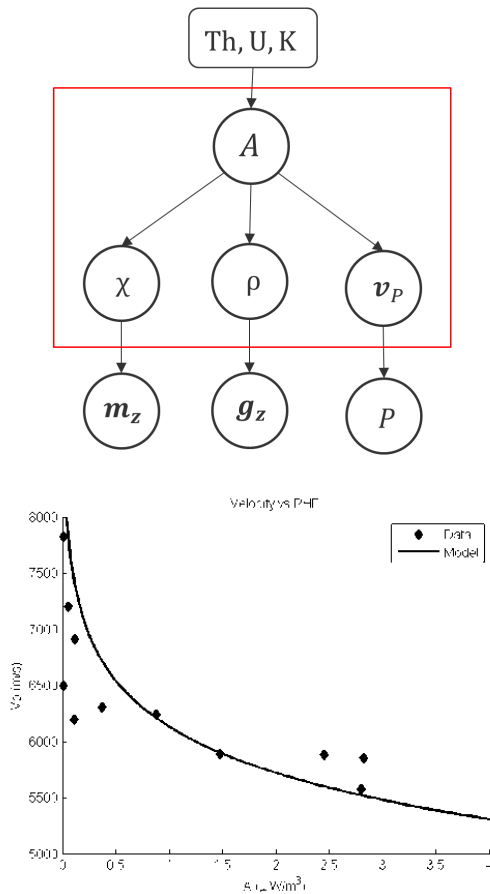
$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i | x_i^{pa})$$



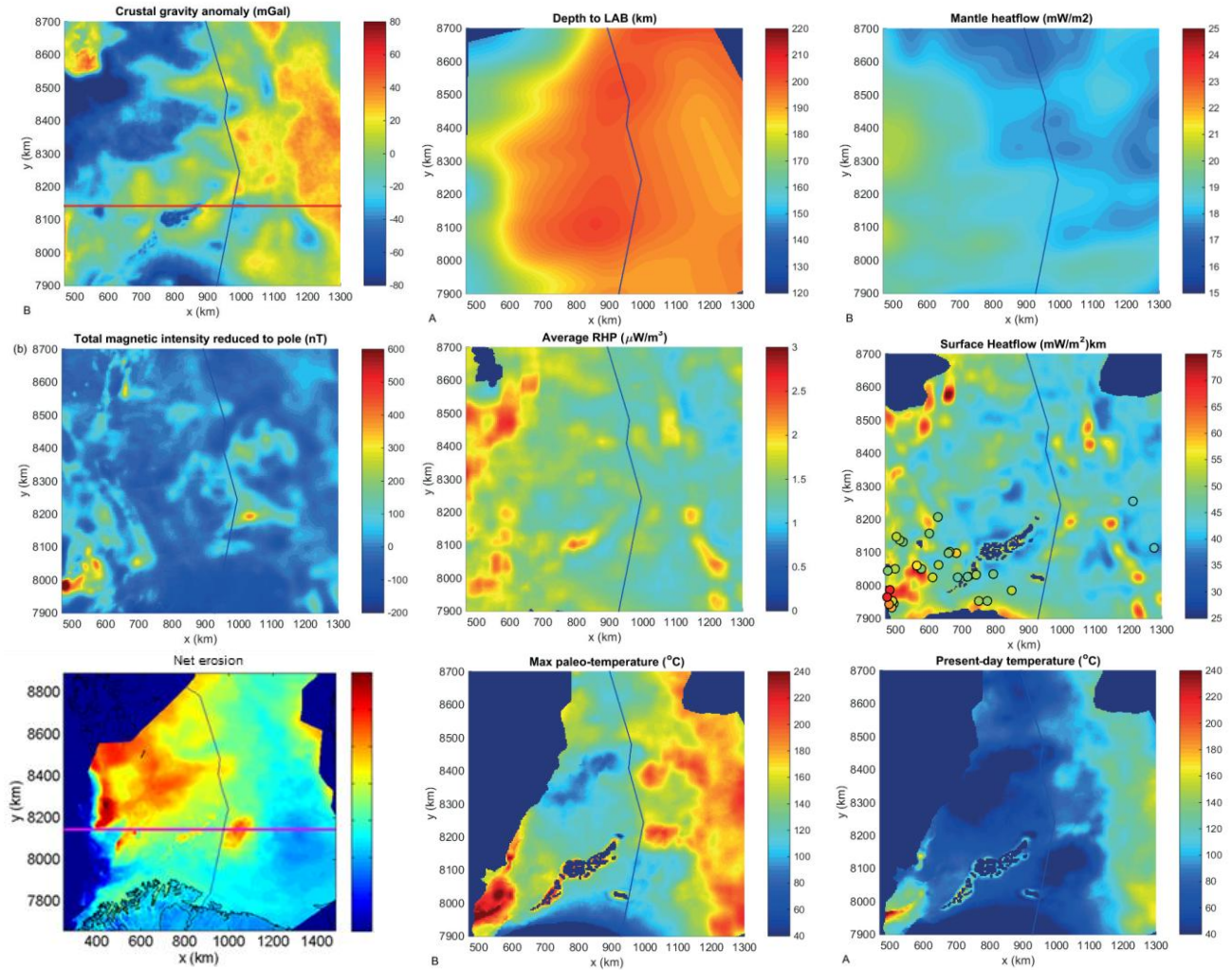
(Martinelli et al., SPE, 2014)

Fig. 3—BN representation of the basin. The Prospects A through E are in the center nodes. Their immediate parents are the observable success factors, and these are again linked through networks. On the left is the trap subnetwork, on the top the producibility (PR) and the reservoir presence (RP) subnetworks, on the right the source abundance (SA) subnetwork, on the bottom the source subnetwork.

Petroleum exploration: RHP and heat flow

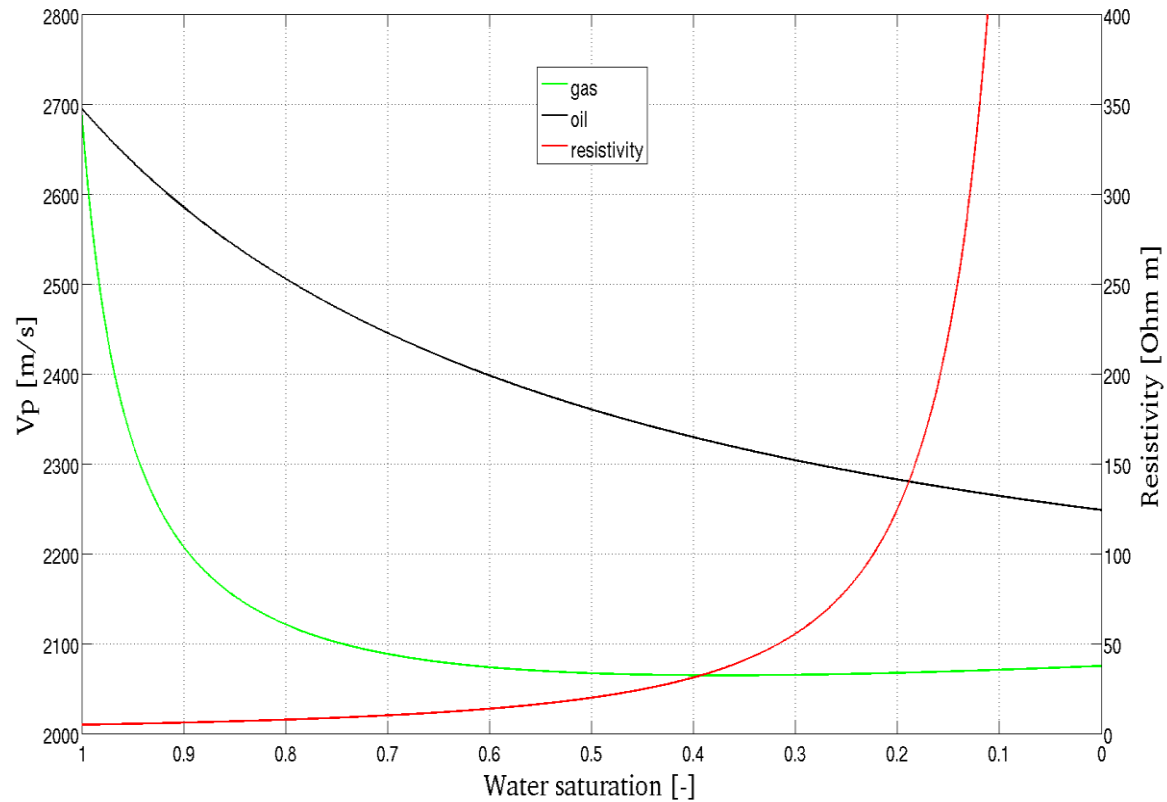


Crustal rock
physics relations

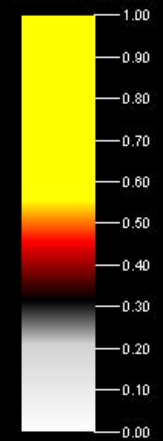


Hokstad et al. (NJG, 2017)

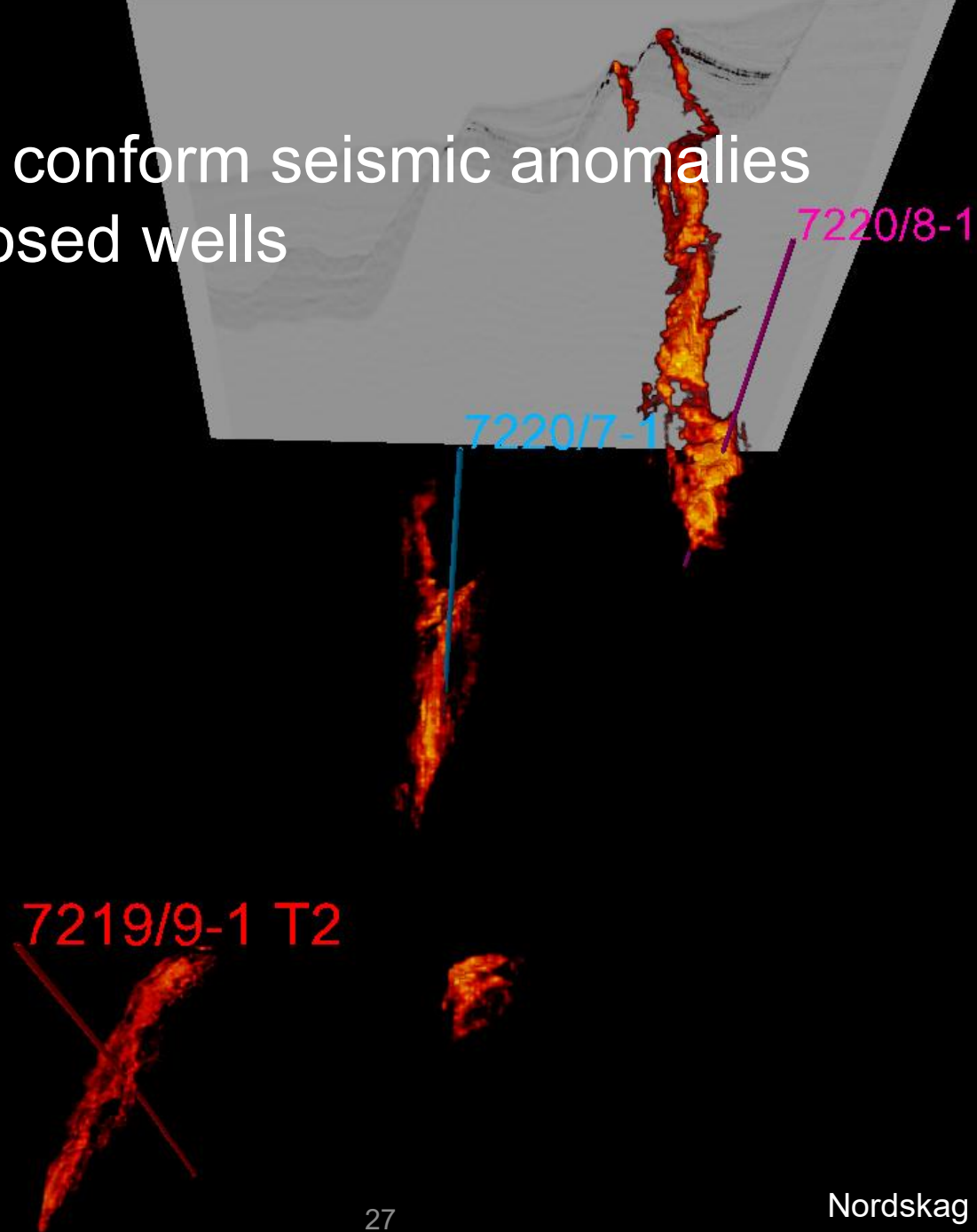
Fluid prediction from seismic and CSEM



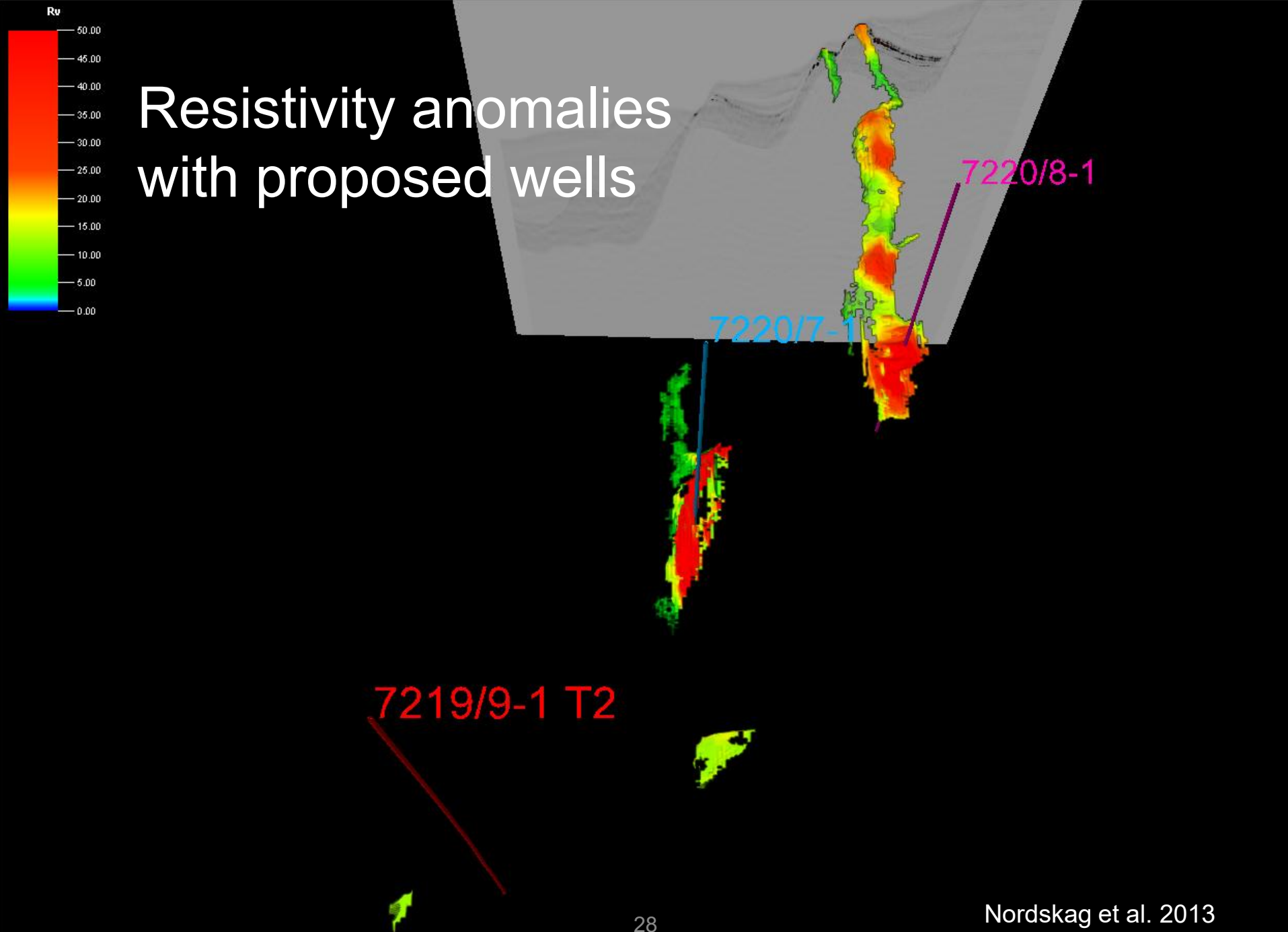
Gassmann vs. Archie equations



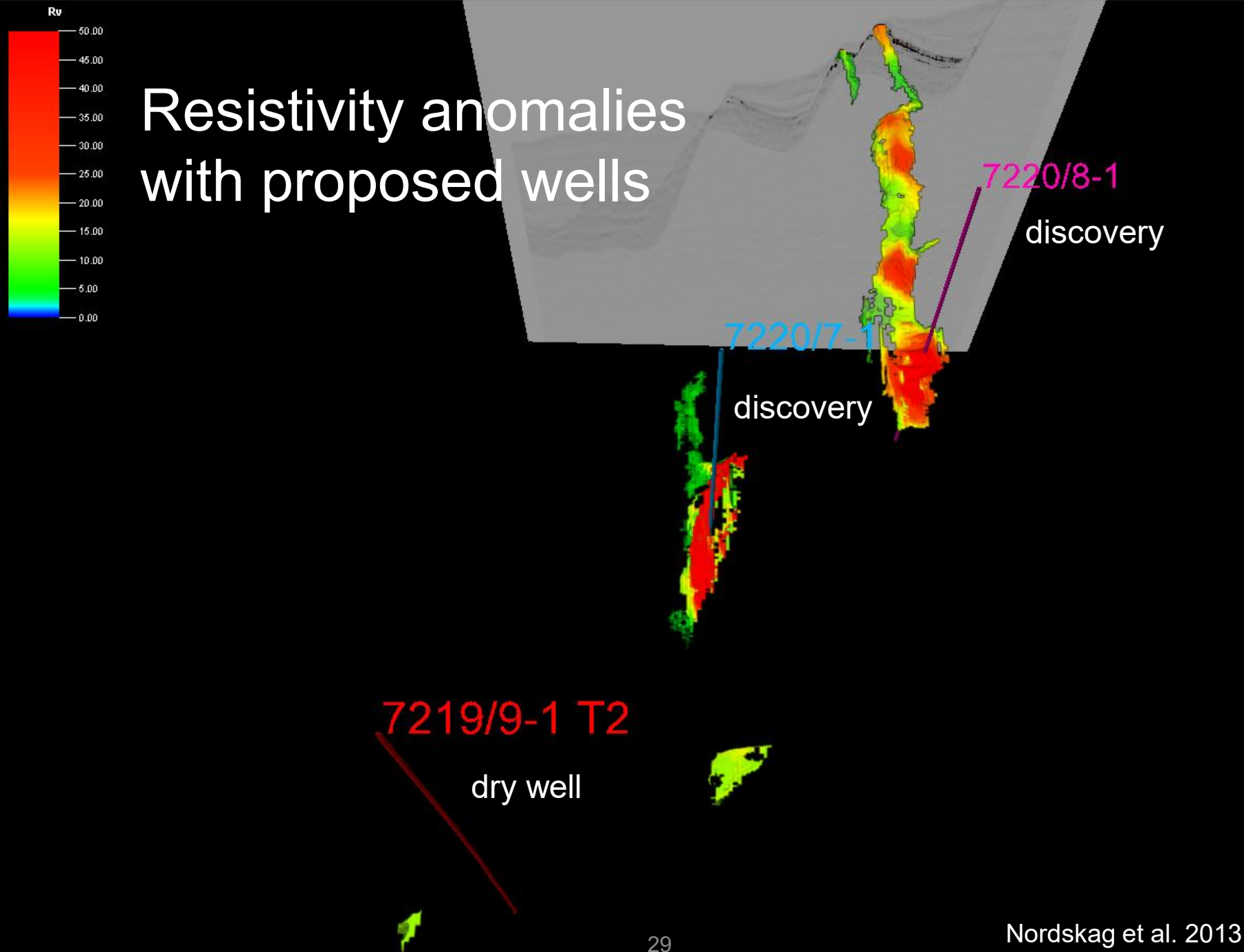
Structural conform seismic anomalies with proposed wells



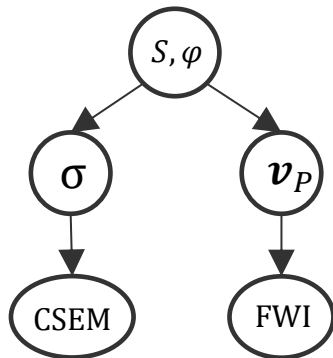
Resistivity anomalies with proposed wells



Resistivity anomalies with proposed wells



Fluid saturation and porosity from resistivity and seismic velocity



Rock physics forward models:

- Gassmann equation

$$v_p = f_v(S, \phi) + e_v$$

- Archie equation

$$R_t = f_R(S, \phi) + e_R$$

- Archie and Gassmann go into the likelihoods

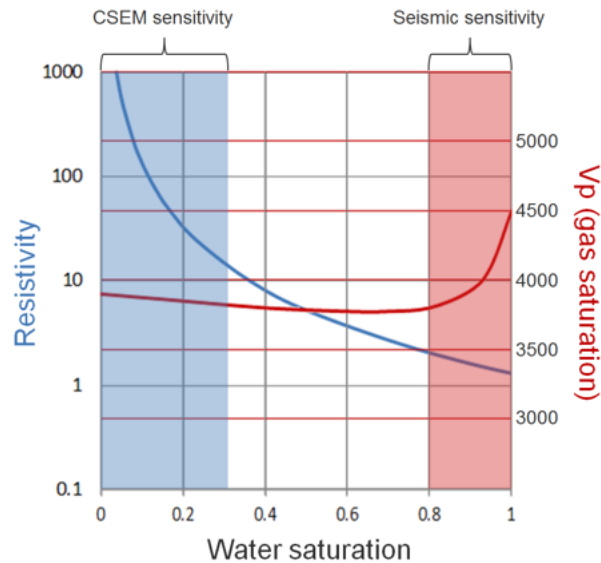
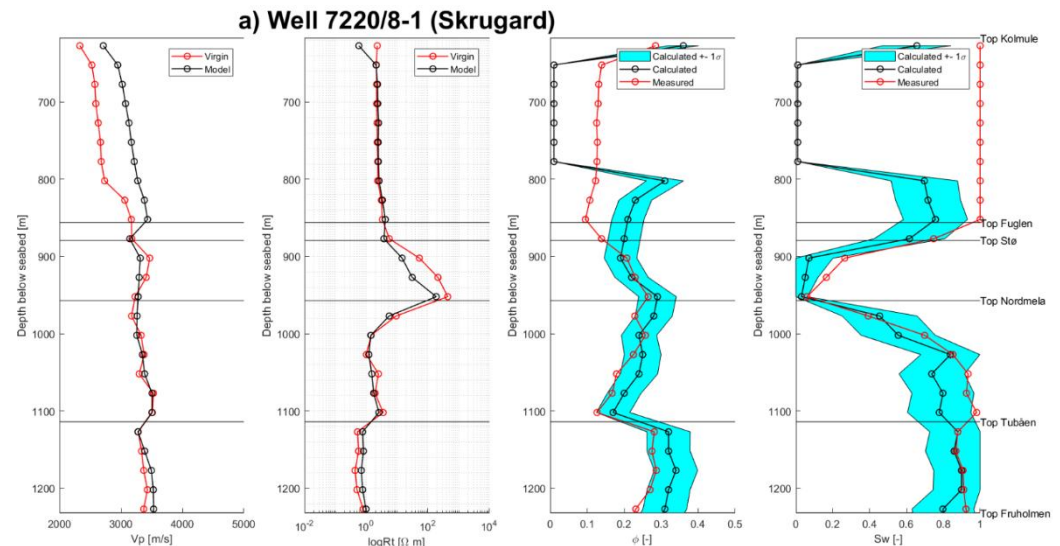


Figure from Constable, 2010



From MSc thesis of Sondre Olsson (2019)

Posterior distribution, mean and variance

Posterior distribution for saturation and porosity

$$p(S, \phi | v_p, R_t; \lambda) \propto p(v_p | S, \phi) p(R_t | S, \phi) p(S, \phi)$$

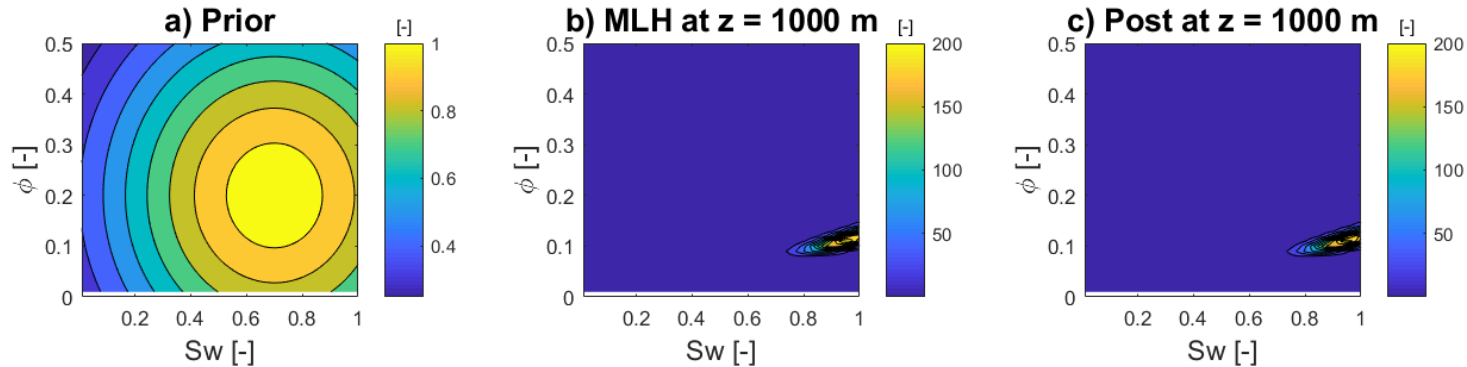
Substituting rock physics models in the likelihoods

$$p(S, \phi | v_p, R_t; \lambda) \propto \underbrace{e^{-\frac{(v_p - f_v)^2}{2\sigma_{ev}^2}} e^{-\frac{(R_t - f_R)^2}{2\sigma_{eR}^2}}}_{\text{Likelihood}} \underbrace{e^{-\frac{(v_p - \mu_v)^2}{2\sigma_v^2}} e^{-\frac{(R_t - \mu_R)^2}{2\sigma_R^2}}}_{\text{Prior}}$$

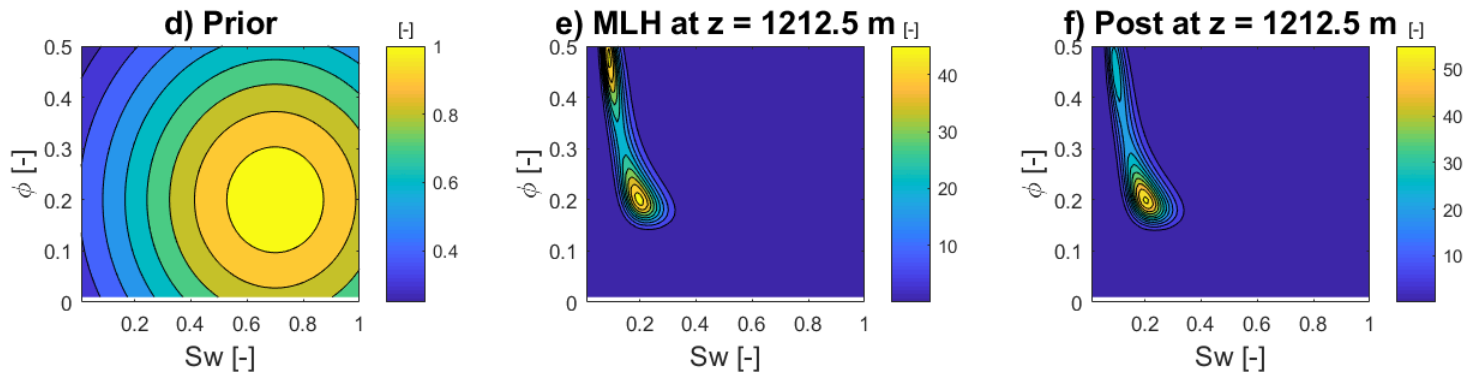
Posterior mean and variance computed from the definitions

Synthetic examples of distributions

Water case

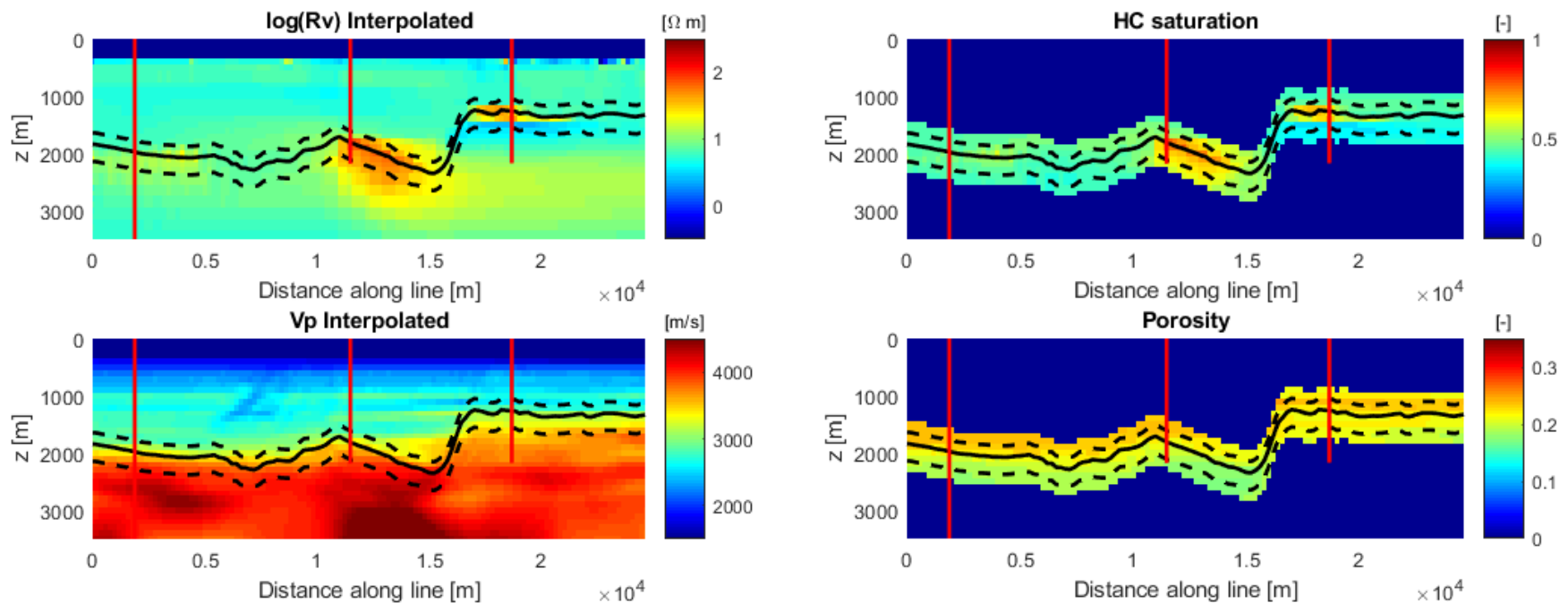


Hydrocarbon case



From MSc thesis of Sondre Olsson (2019)

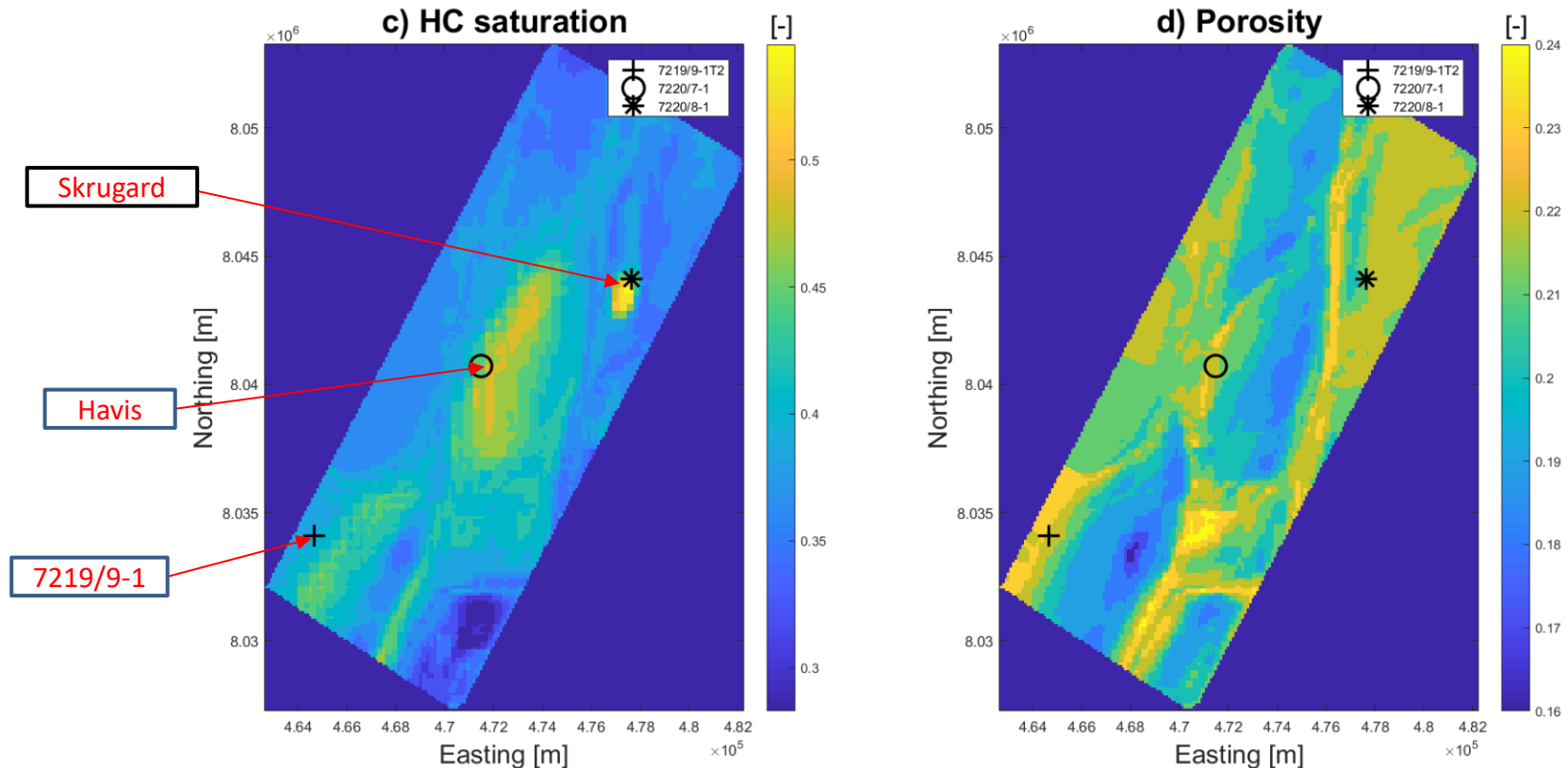
Johan Castberg area – random line



From MSc thesis of Sondre Olsson (2019)

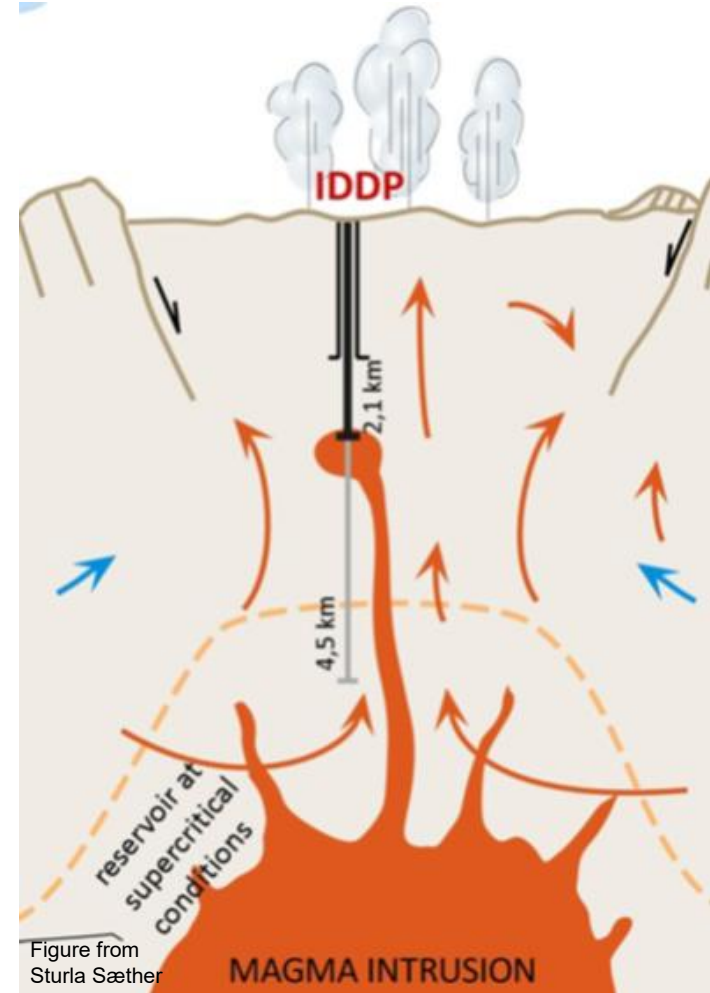
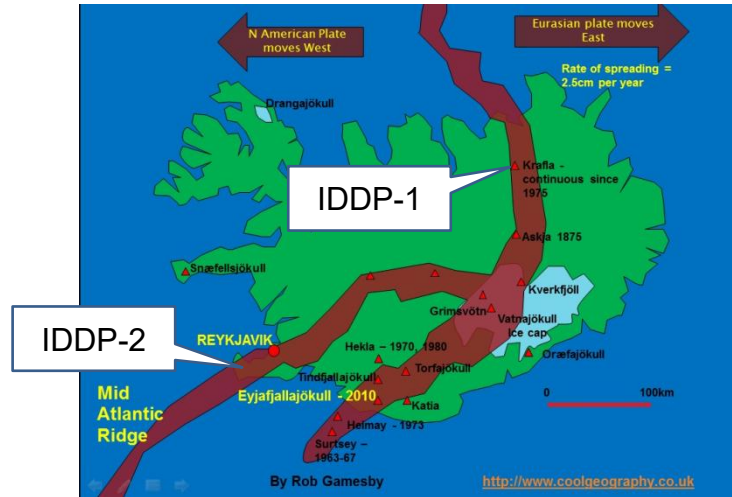
Johan Castberg area – map view

Realgrunnen Gp (Jurassic reservoir)



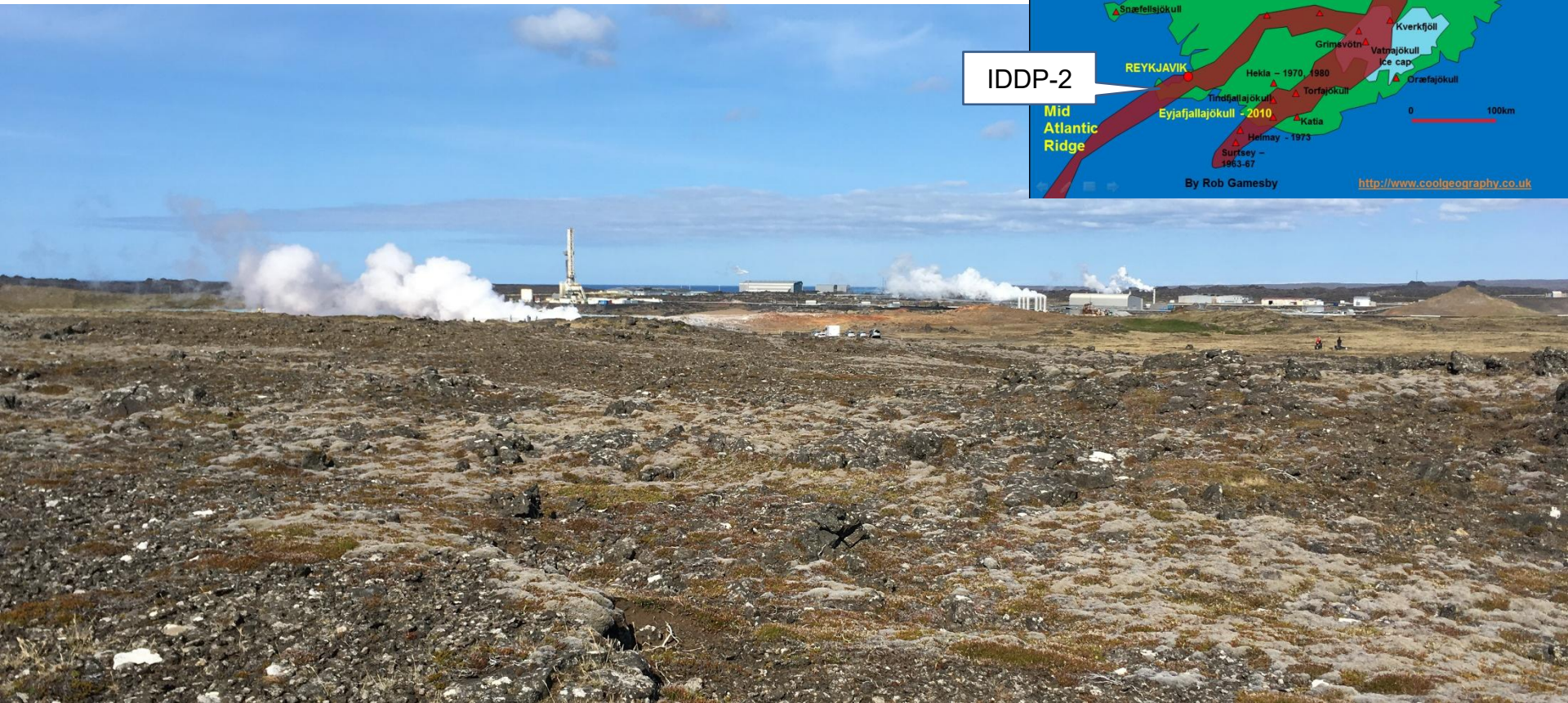
From MSc thesis of Sondre Olsson (2019)

Geothermal exploration: Subsurface temperature

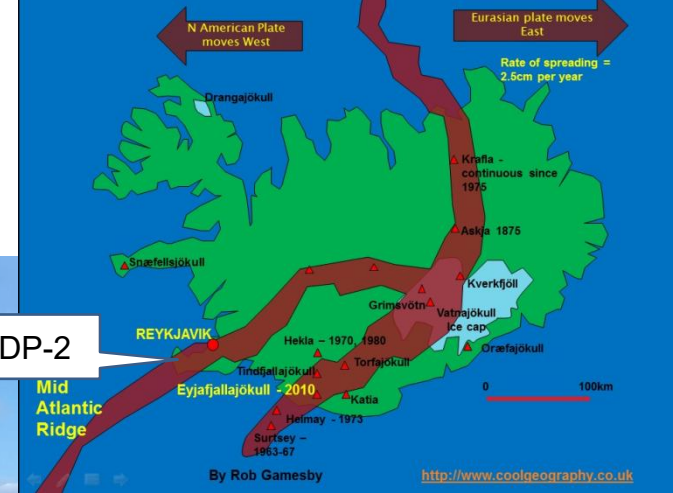


IDDP-1: Drilled into an unexpected
magma chamber at 2.1km depth

Geothermal exploration



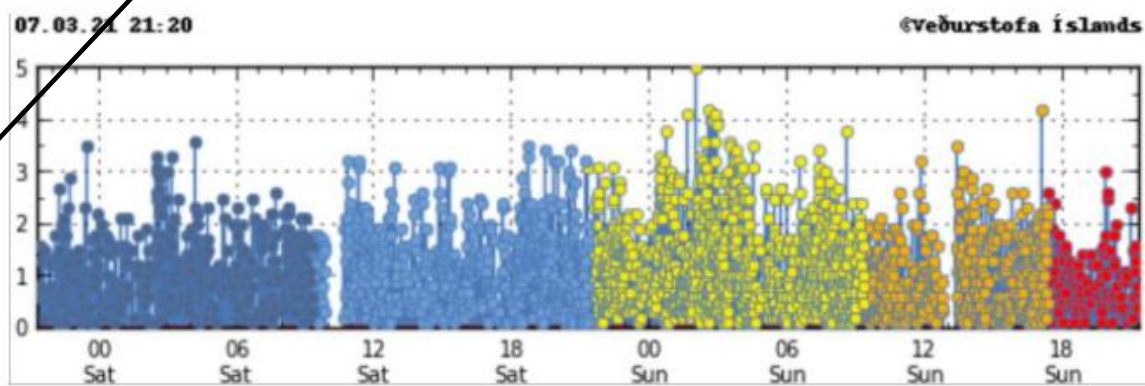
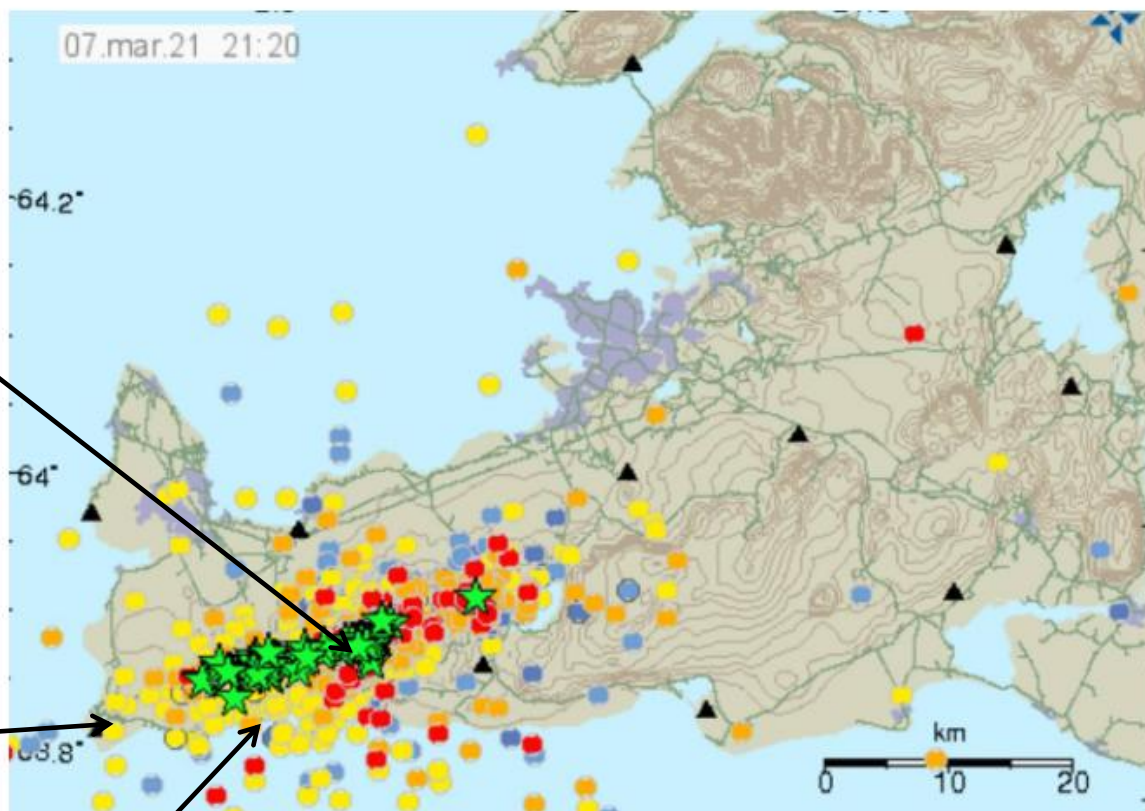
IDDP-2



- Reykjavanes geothermal area
- ~30 conventional production and injection wells

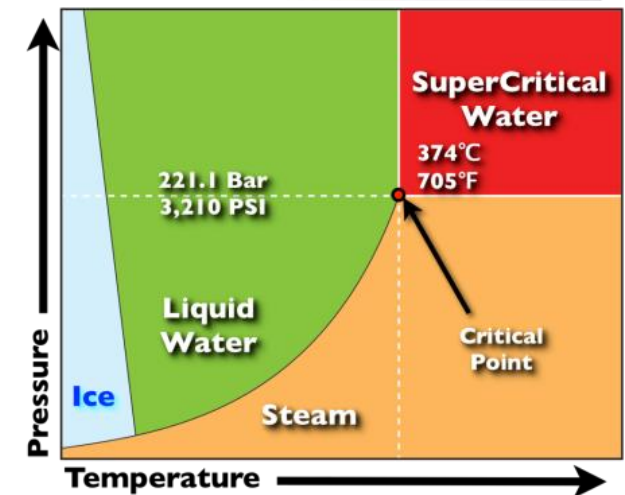
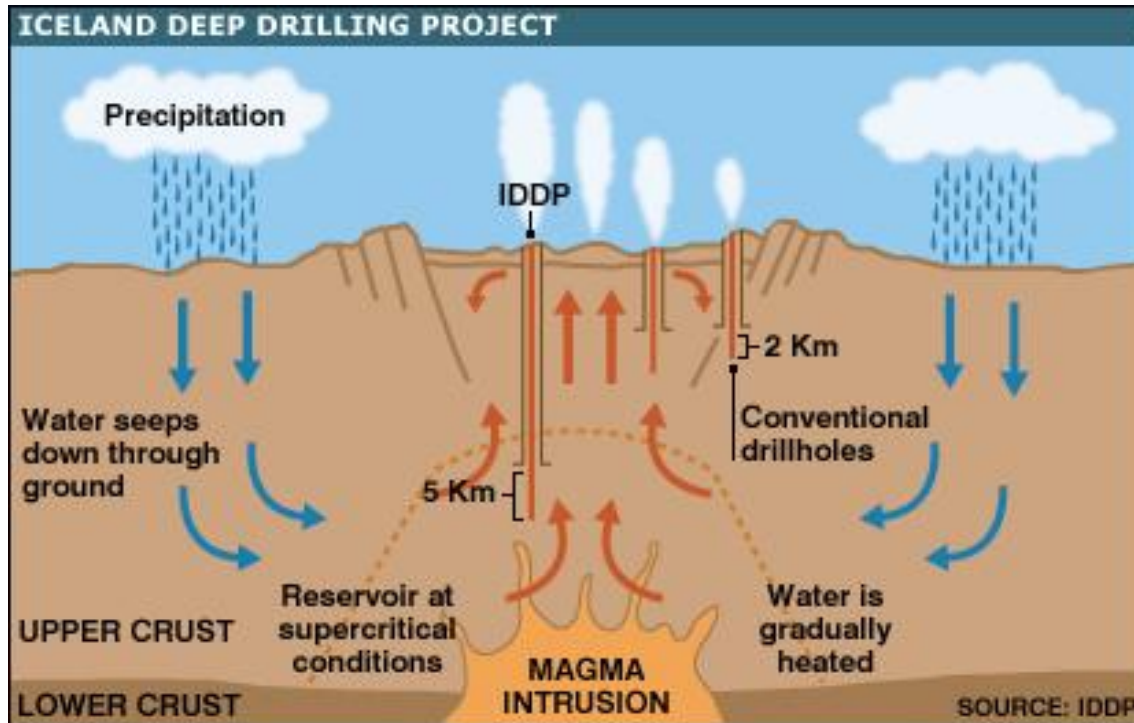


2015-02-24



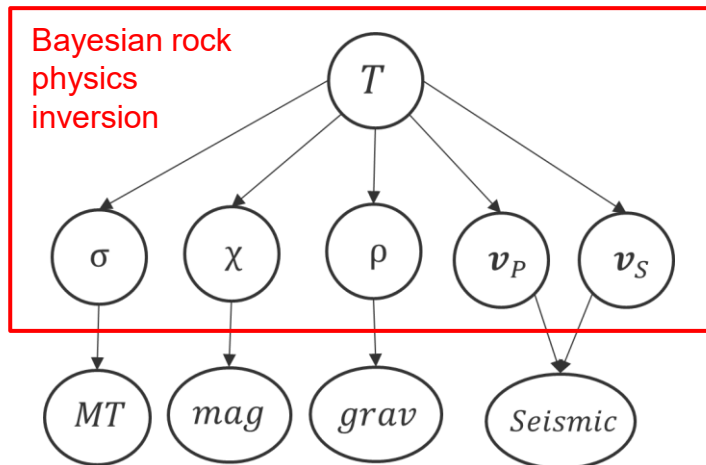
Classification: Open

Exploring for supercritical water



- To find supercritical water we need to drill deeper
- How deep?

MGI-GT Bayesian inversion



$$p(T|m_1, \dots, m_n; \phi) = \prod_{i=1}^n \frac{p(m_i|T; \phi)}{p(m_i; \phi)} p(T)$$

Posterior distribution for temperature, given resistivity and density:

$$p(T|R_h, \rho; \phi) \propto p(R_h|T; \phi) p(\rho|T; \phi) p(T)$$

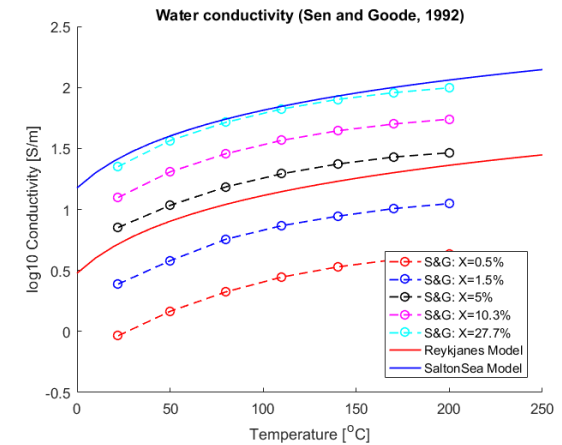
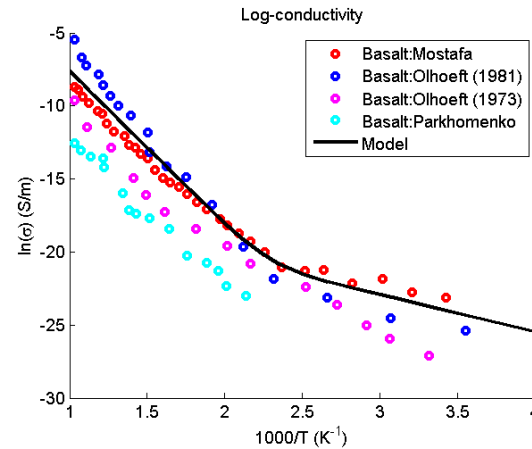
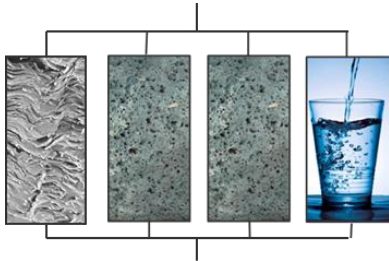
Posterior mean temperature and variance:

$$\mu_{T|R_h, \rho} = \int T p(T|R_h, \rho; \phi) dT$$

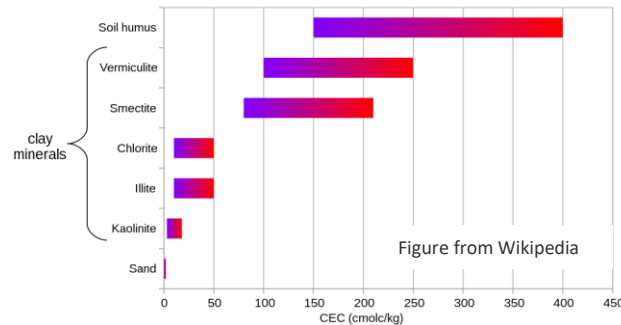
$$\sigma_{T|R_h, \rho}^2 = \int (T - \mu_{T|R_h, \rho})^2 p(T|R_h, \rho; \phi) dT$$

- Two-stage approach:
 1. Geophysical inversion
 2. **Rock-physics inversion**

Electric conductivity (resistivity)



- Parallell coupling with
 - Igneous rock (basalt)
 - Pore-water (salts)
 - Clay (cation exchange in smectites)

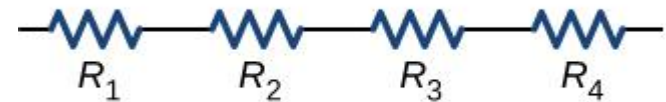


- Cation exchange capacity:
- Waxman-Smiths equation

Resistors in series and parallel

Series:

$$R = R_1 + R_2 + R_3 + R_4$$

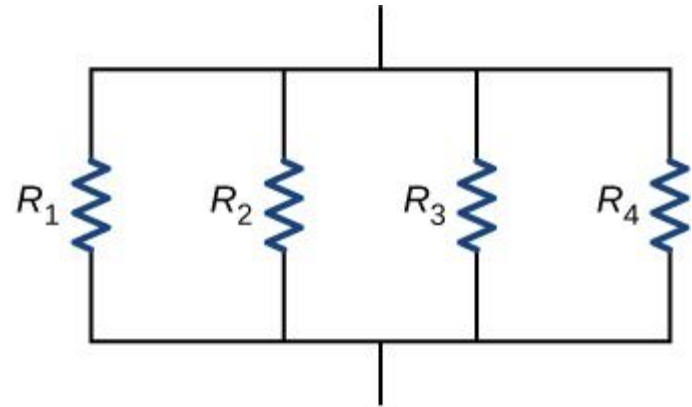


Parallel:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

$$S = S_1 + S_2 + S_3 + S_4$$

$$S = \frac{1}{R} = \text{conductance}$$



Ohm's law: $U = RI$

Density vs temperature

Matrix density (Hacker et al., 2003):

$$\rho_m(T) = \rho_0 e^{-\Phi}$$

$$\Phi(T) = \ln \frac{V(T)}{V_0} = \int_{T_0}^T \alpha(T') dT'$$

$$\alpha(T) = \alpha_0 (1 - 10\sqrt{T})$$

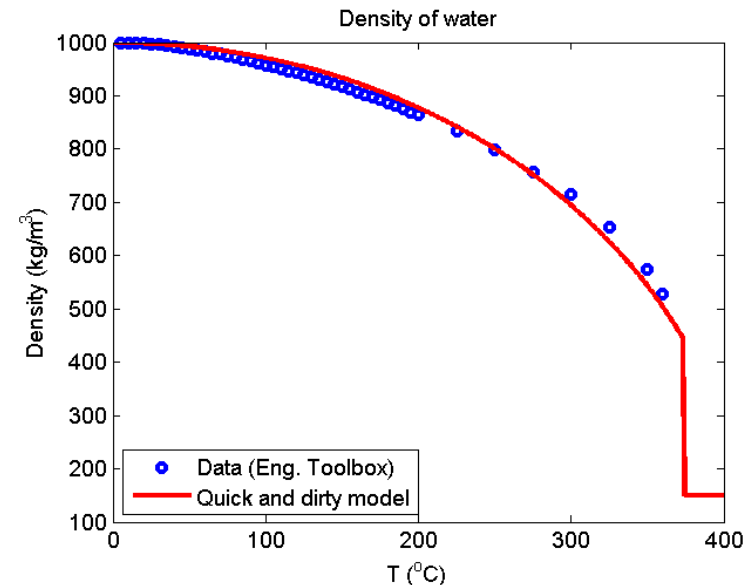
Water density:

$$\rho_w(T) = \sqrt{\rho_0^2 - c^2(T - T_0)^2}$$

$$c = 2.4$$

Bulk density:

$$\rho(T) = (1 - \phi - V_{cl})\rho_m(T) + \phi\rho_w(T) + V_{cl}\rho_c$$

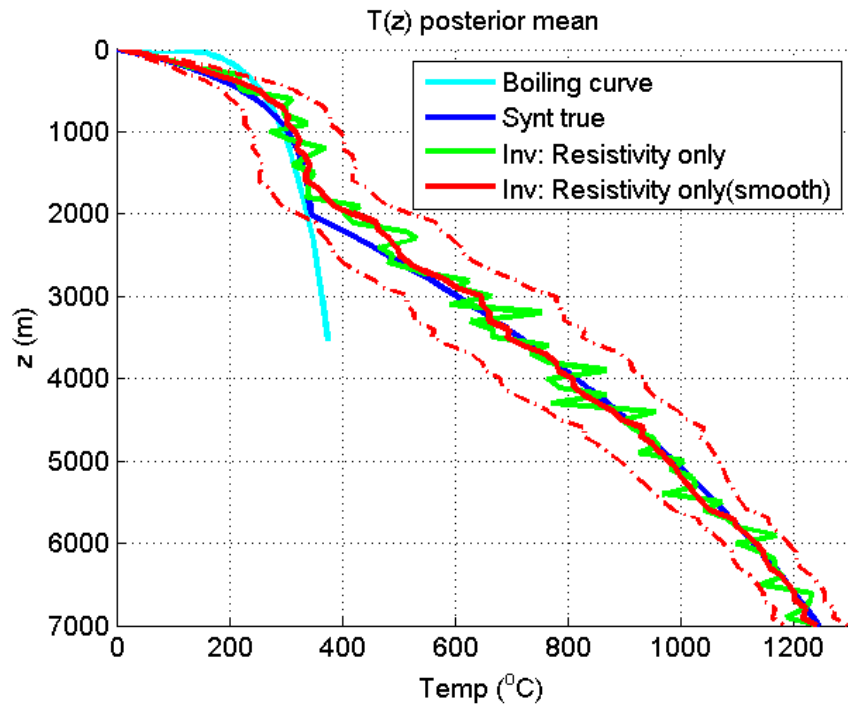


Hydrostatic pressure:

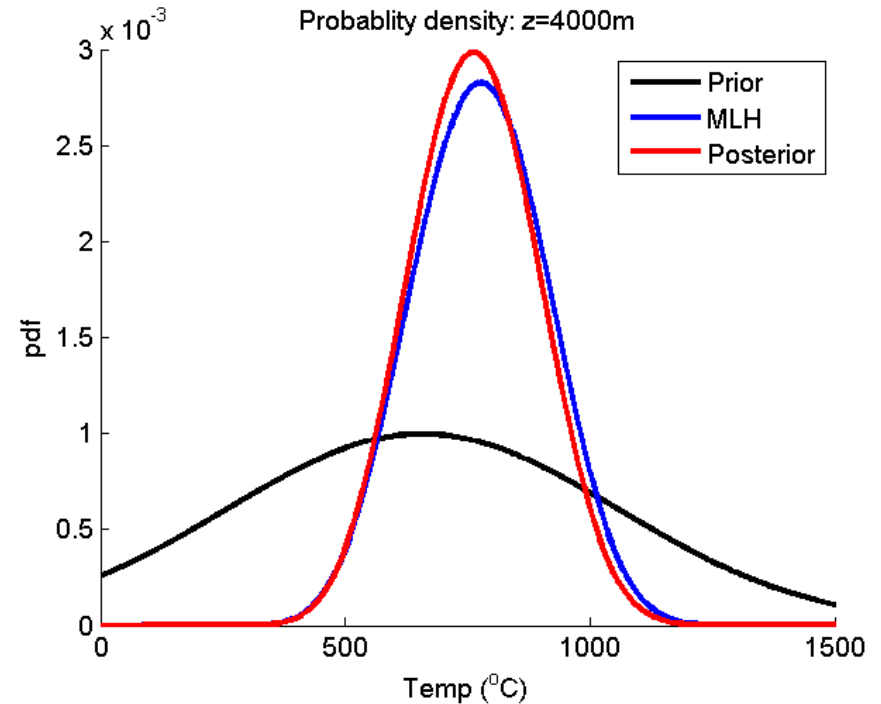
$$P(z) = P_0 + g \int_0^z \rho_w(T(z')) dz'$$

$$\rho_w(T) = \sqrt{\rho_0^2 - c^2(T - T_0)^2}$$

Posterior $T(z)$ – resistivity inversion



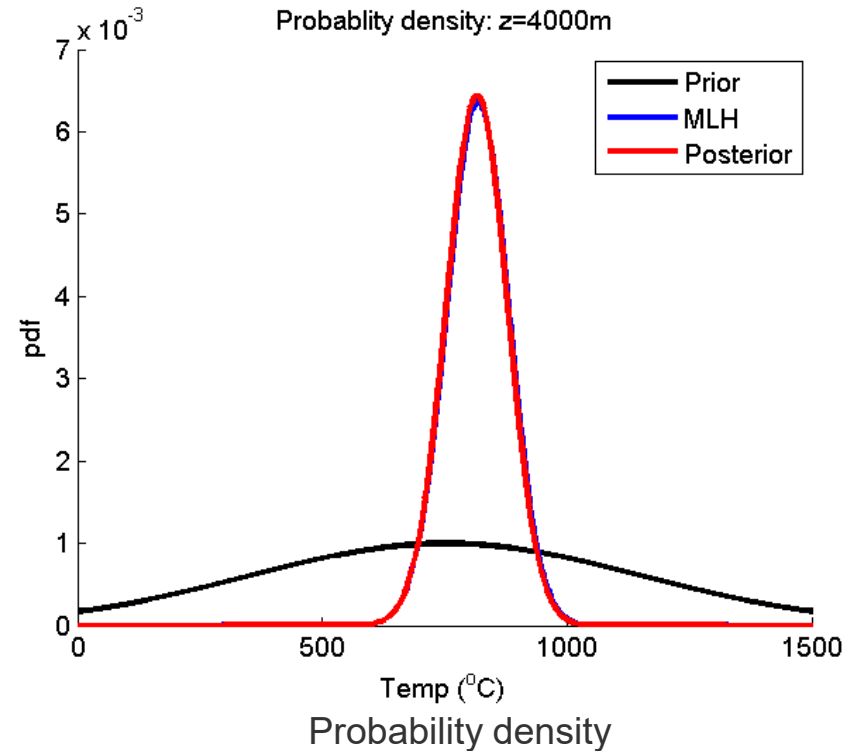
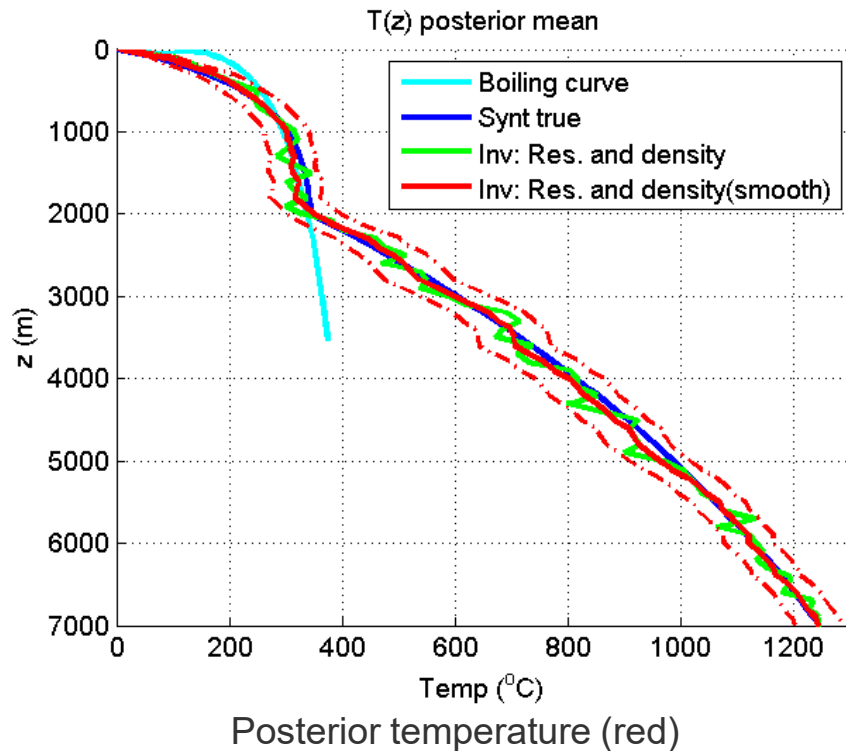
Posterior temperature (red)



Probability density

Synthetic model example

Posterior $T(z)$ – resistivity and density inversion



Synthetic model example

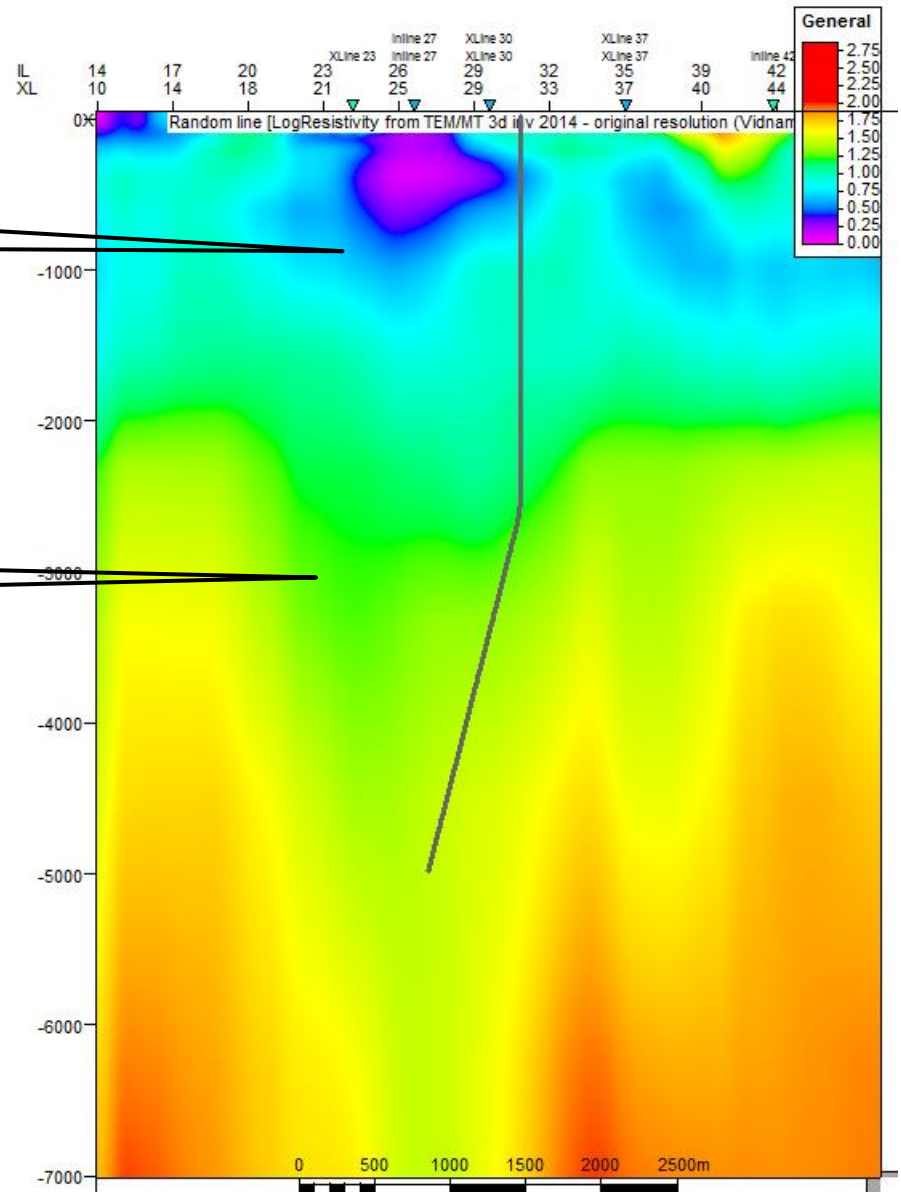
Reykjanes:

3D MT inversion

Low resistivity => Low temperature
(smectite diagenesis)

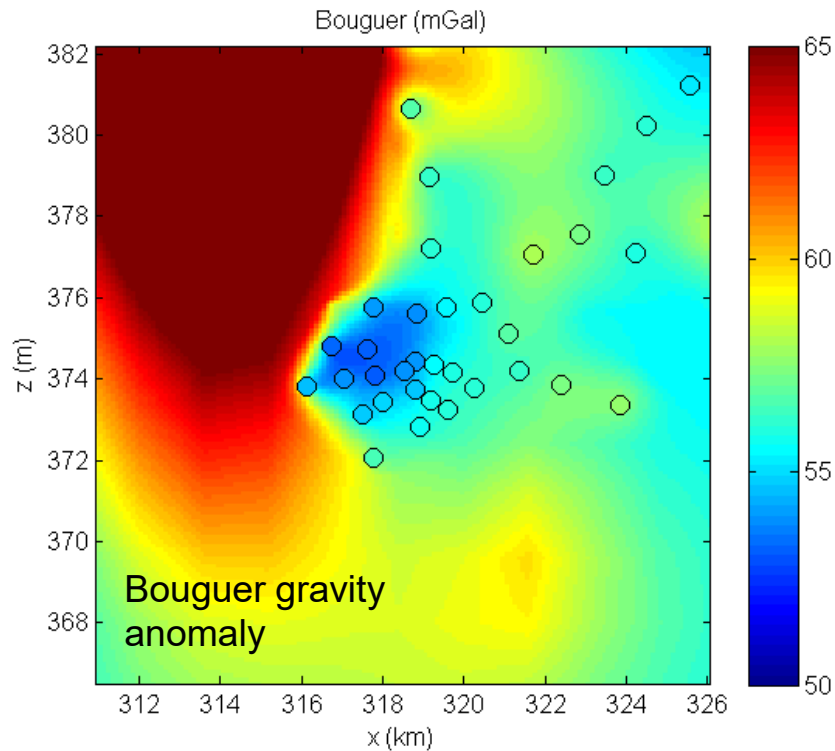
Low resistivity => elevated temperature

- 3D MT + TEM inversion by ISOR (Ragna Karlsdóttir)
- MT inversion code by Siripunvaraporn (2005); minimum norm method

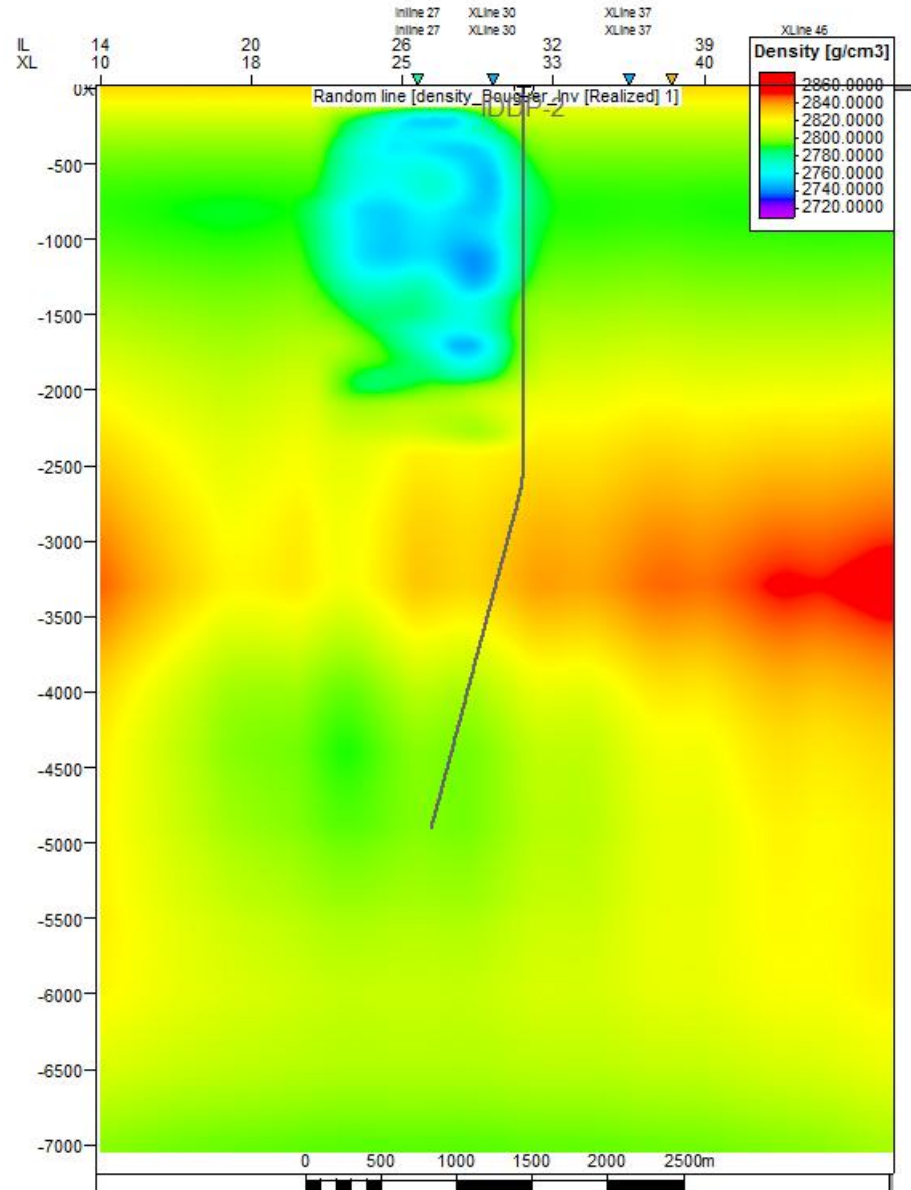


Reykjanes

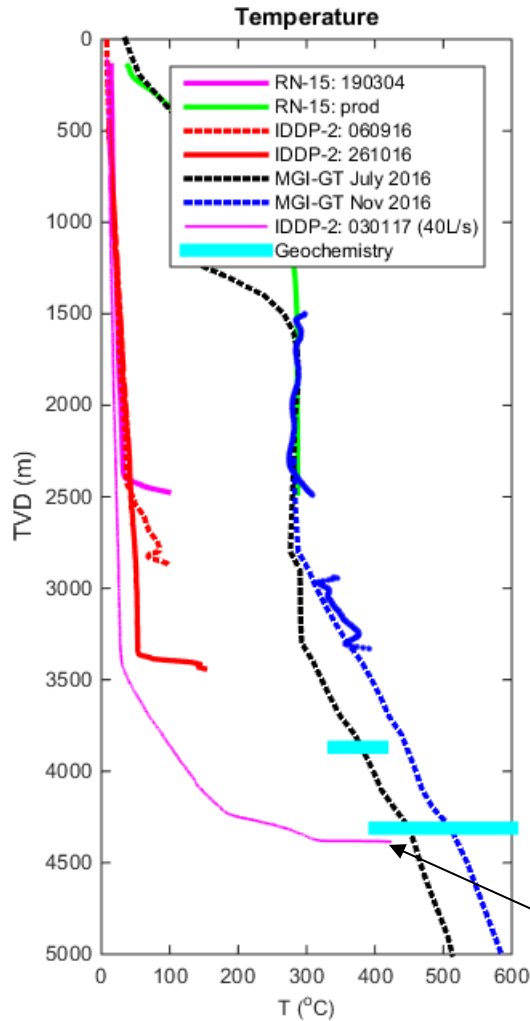
3D gravity inversion



- 3D gravity inversion:
Marquardt-Levenberg scheme



Temperature

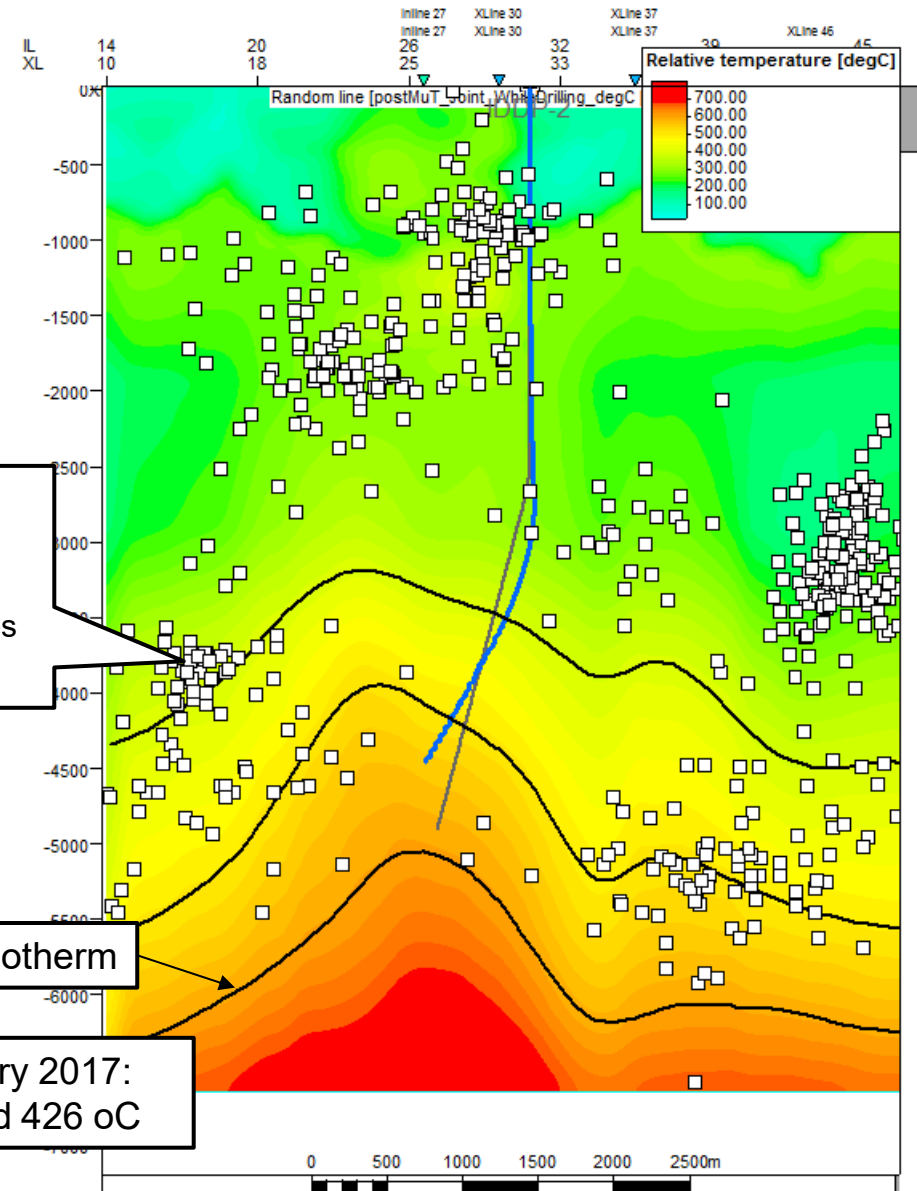


□ Earthquake locations

No earthquakes in ductile zone

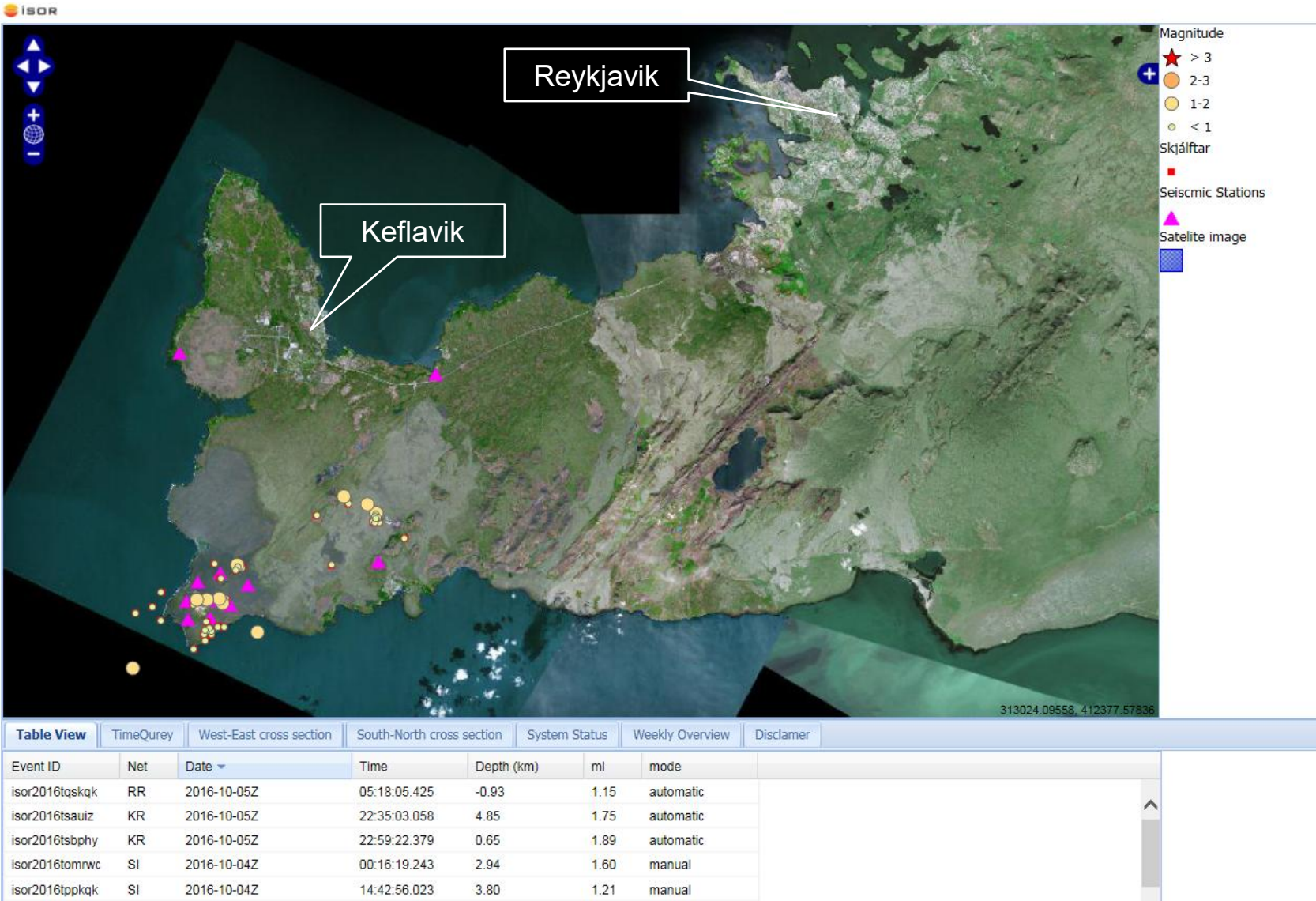
600 °C isotherm

3. January 2017:
Measured 426 °C

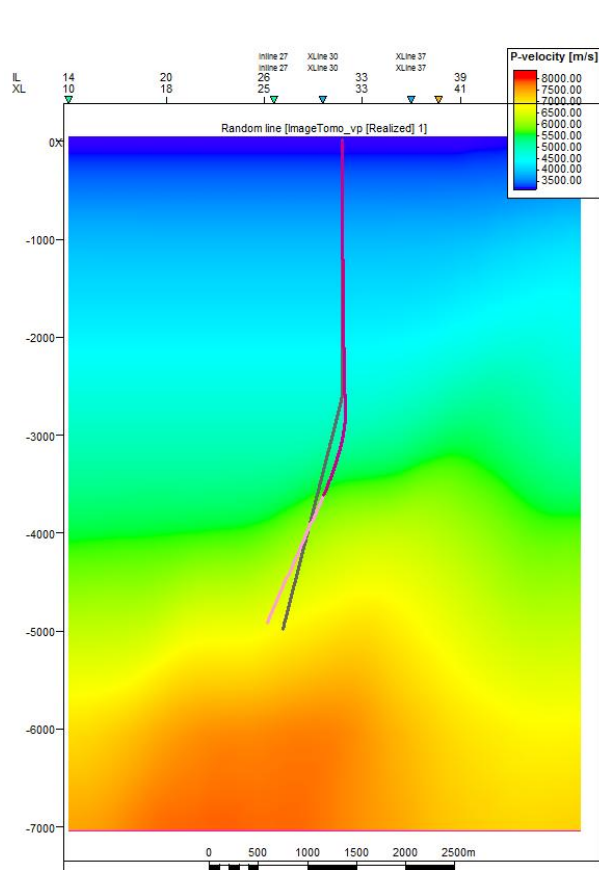


Reykjanes - Automatic Mirco Earthquake Locations

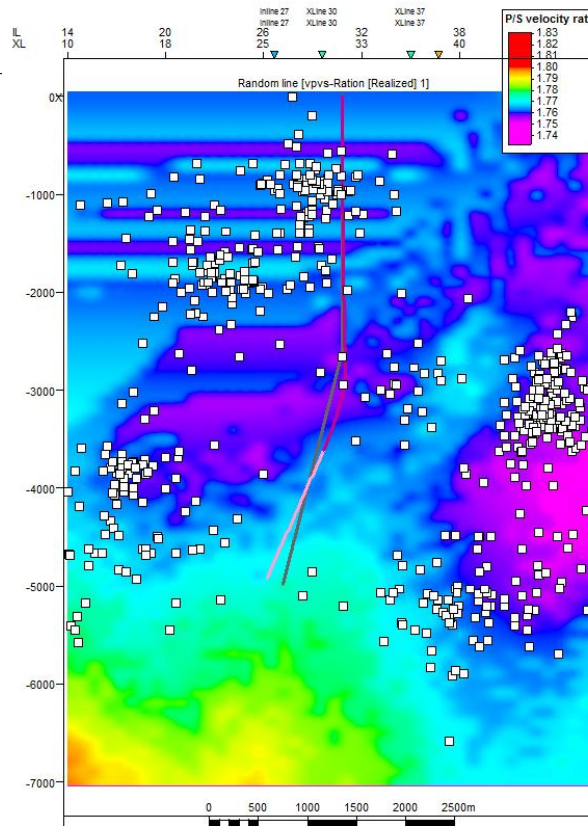
Initial view shows data from the last 7 days



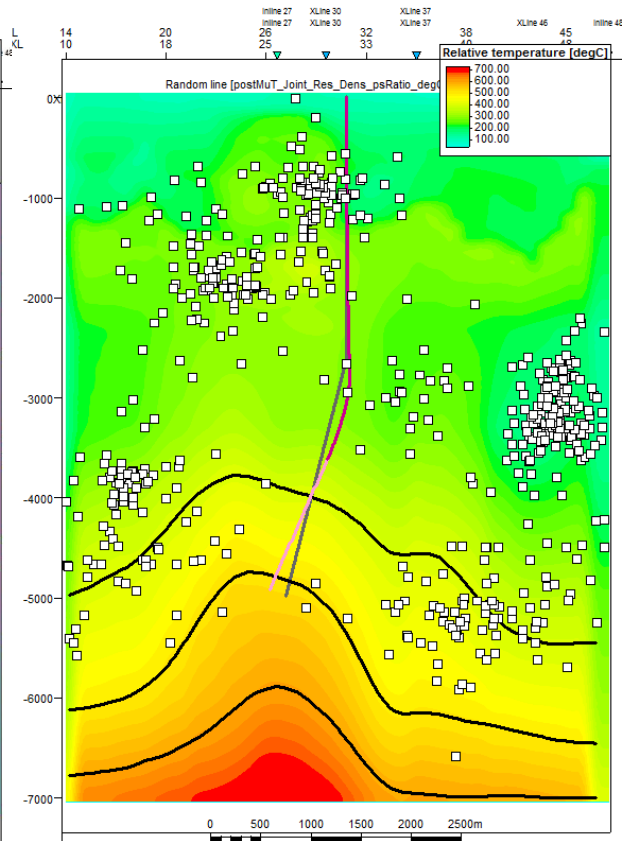
Including seismic velocity from earth quakes



P-wave velocity

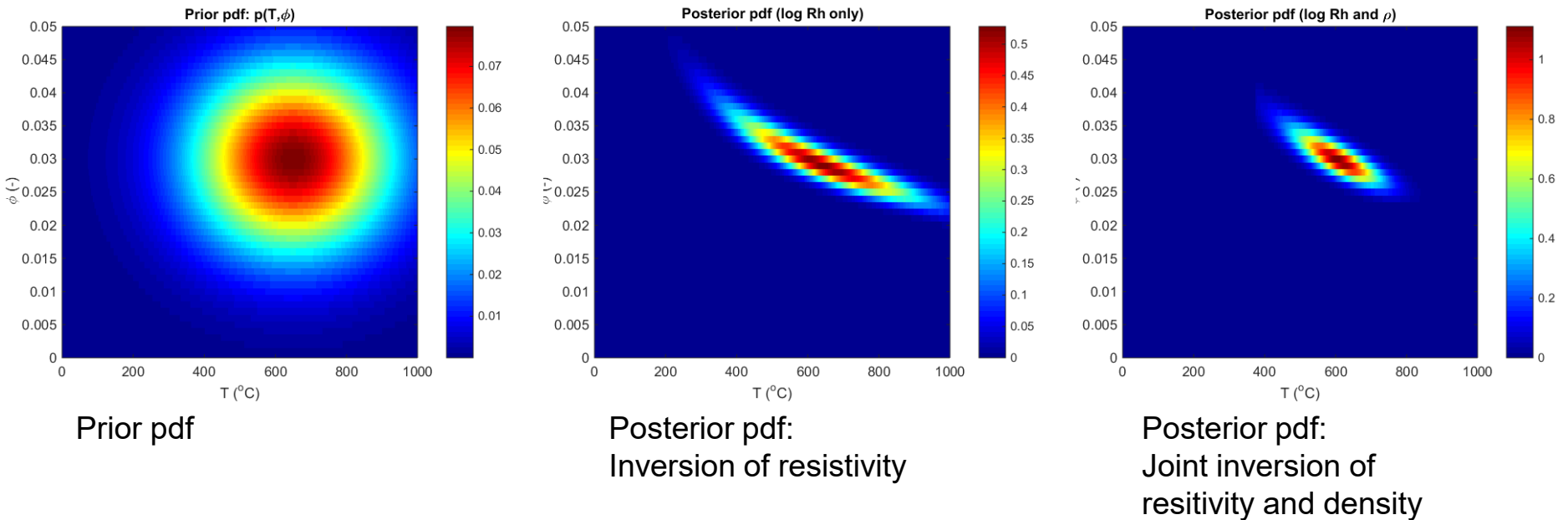


Vp/vs-ratio



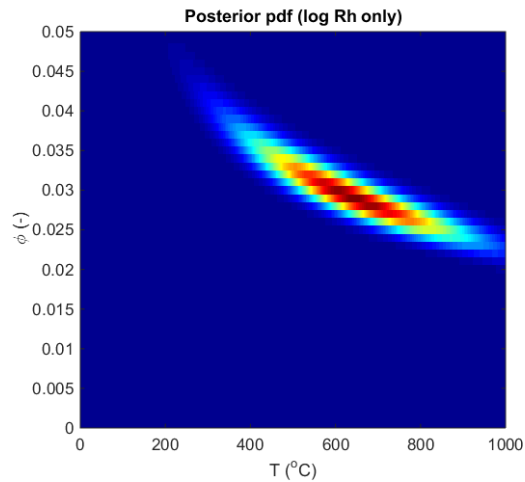
Joint inversion of resistivity,
density and vp/vs-ratio

Illposedness: Inversion for temperature and porosity



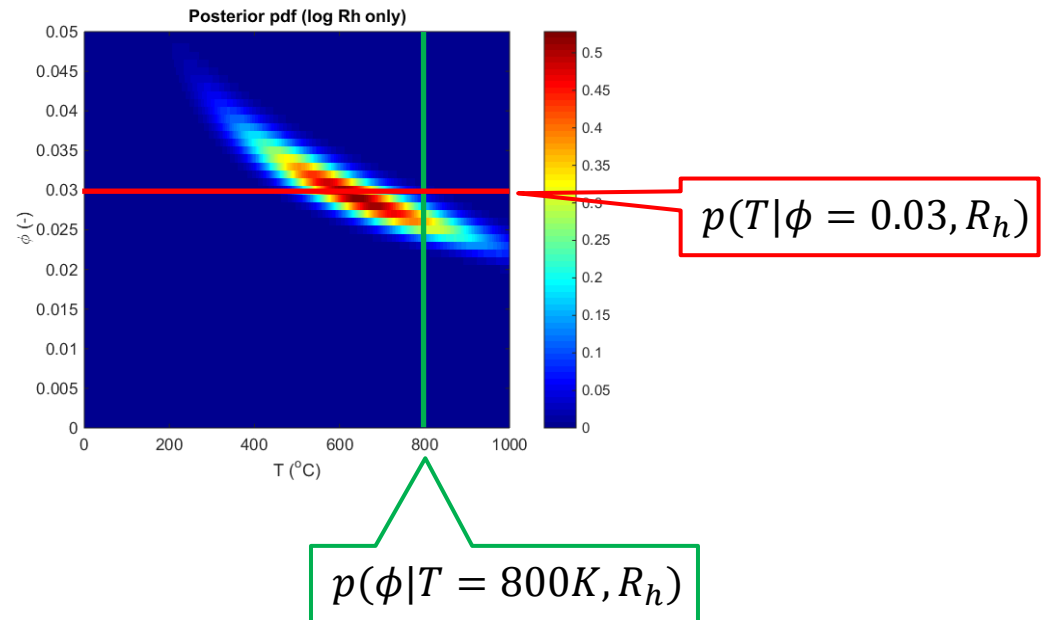
$$p(T, \phi | m_1, \dots, m_n) = C \prod_{i=1}^n p(m_i | T, \phi) p(T, \phi)$$

What is a conditional distribution?

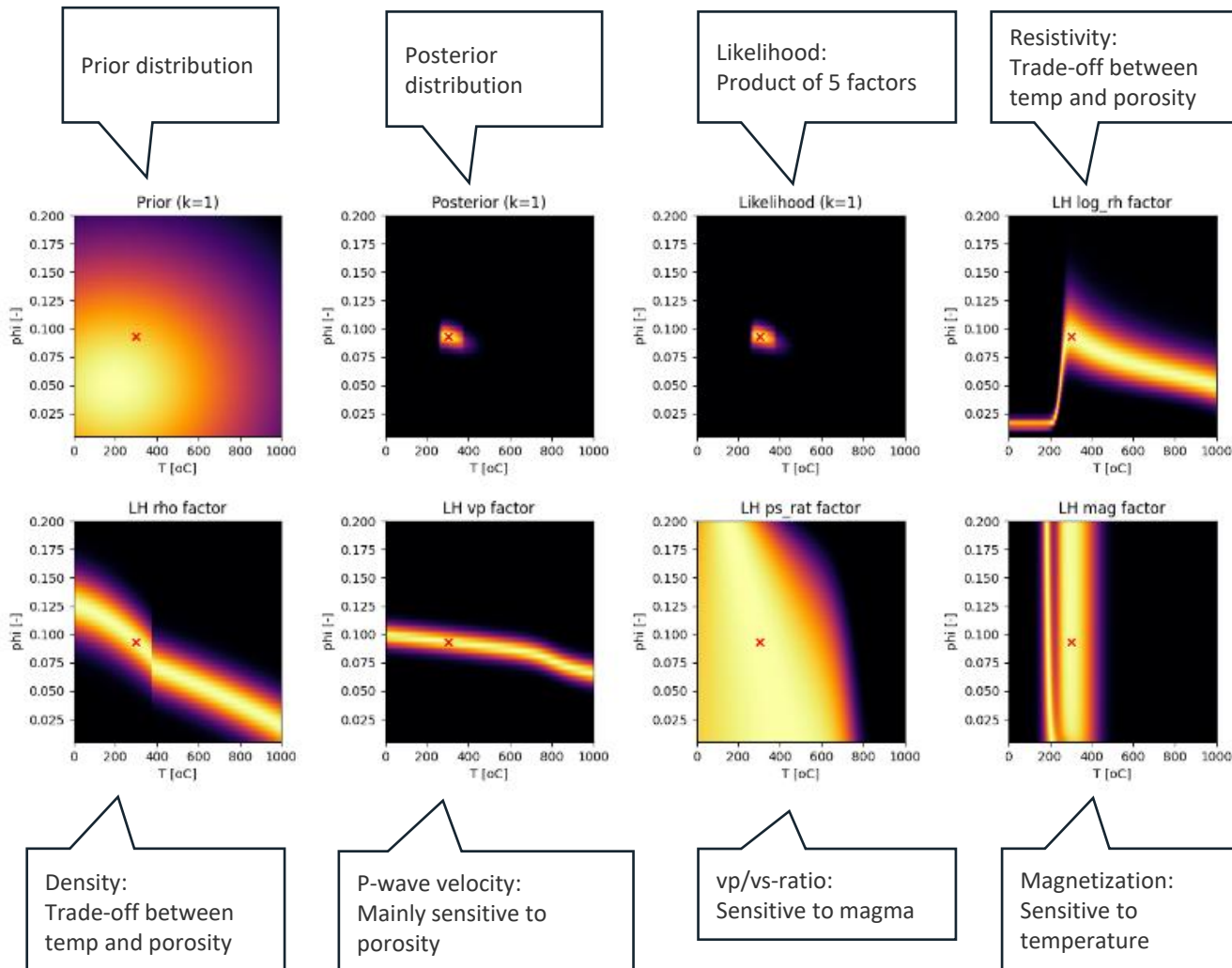


$$p(T, \phi | R_h)$$

Joint distribution for T
and ϕ given R_h



Bivariate MGI: Synthetic geothermal example and sensitivity



Joint distributions for temperature and porosity

- Posterior
- Likelihoods
- Prior

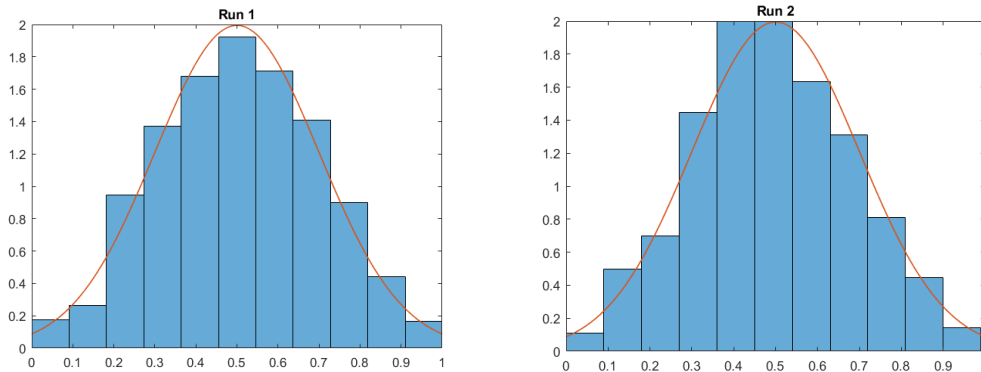
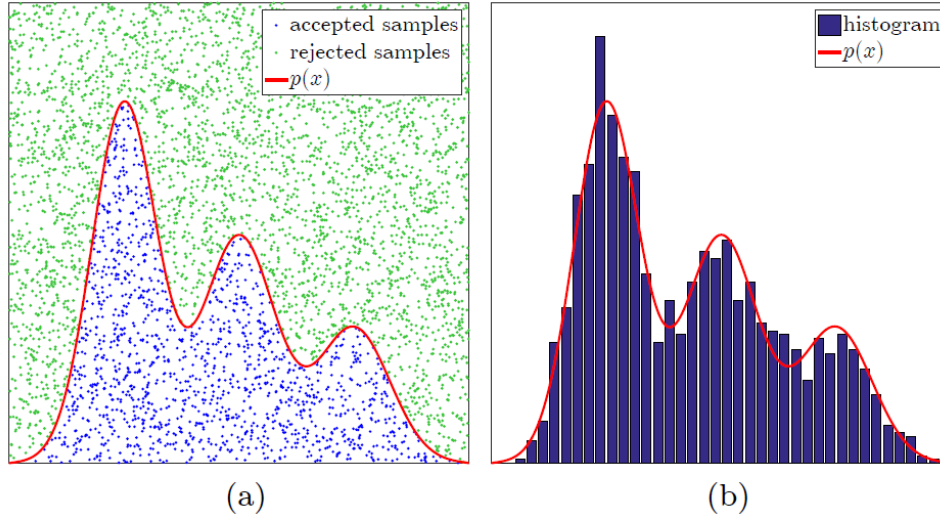
Test model parameters:

```

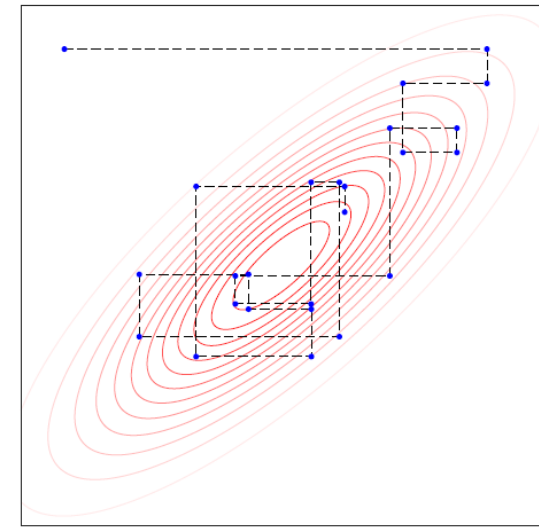
mgi_test_mod = {
    'T': 574.09,
    'phi': 0.09318952,
    'log_rho': 1.53518958,
    'rho': 2690.74510624,
    'vp': 4337.63930387,
    'ps_rat': 1.8296102,
    'mag': 8.55688081
}
    
```

Accept/Reject sampler

Gibbs sampler



Test of the accept/reject sampler on a Gaussian



(c) SC Gibbs sampler

Gibbs sampling of $\mathbf{x} = (x_1, x_2)$ from $p(x_1, x_2)$

Sample $x_1^{(1)}$ from $p(x_1^{(1)} | x_2^{(0)})$

Sample $x_2^{(1)}$ from $p(x_2^{(1)} | x_1^{(1)})$

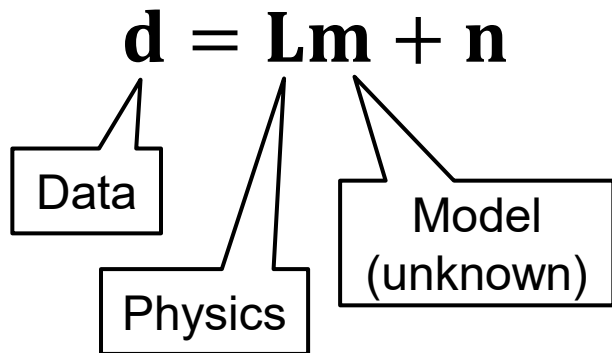
Sample $x_1^{(n)}$ from $p(x_1^{(n)} | x_2^{(n-1)})$

Sample $x_2^{(n)}$ from $p(x_2^{(n)} | x_1^{(n)})$

Need to know $g(x_1, x_2) = Cp(x_1, x_2)$

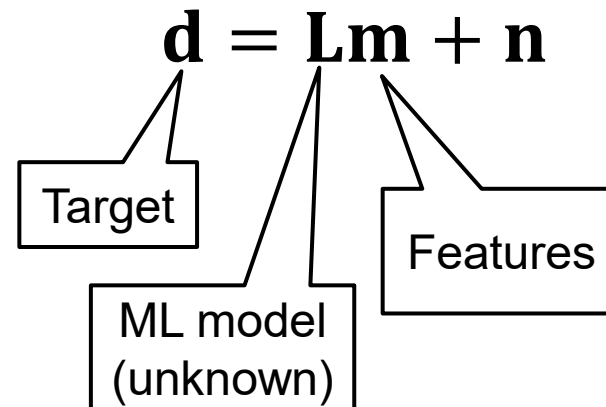
Linear regression: Inversion vs. ML

Linear inversion



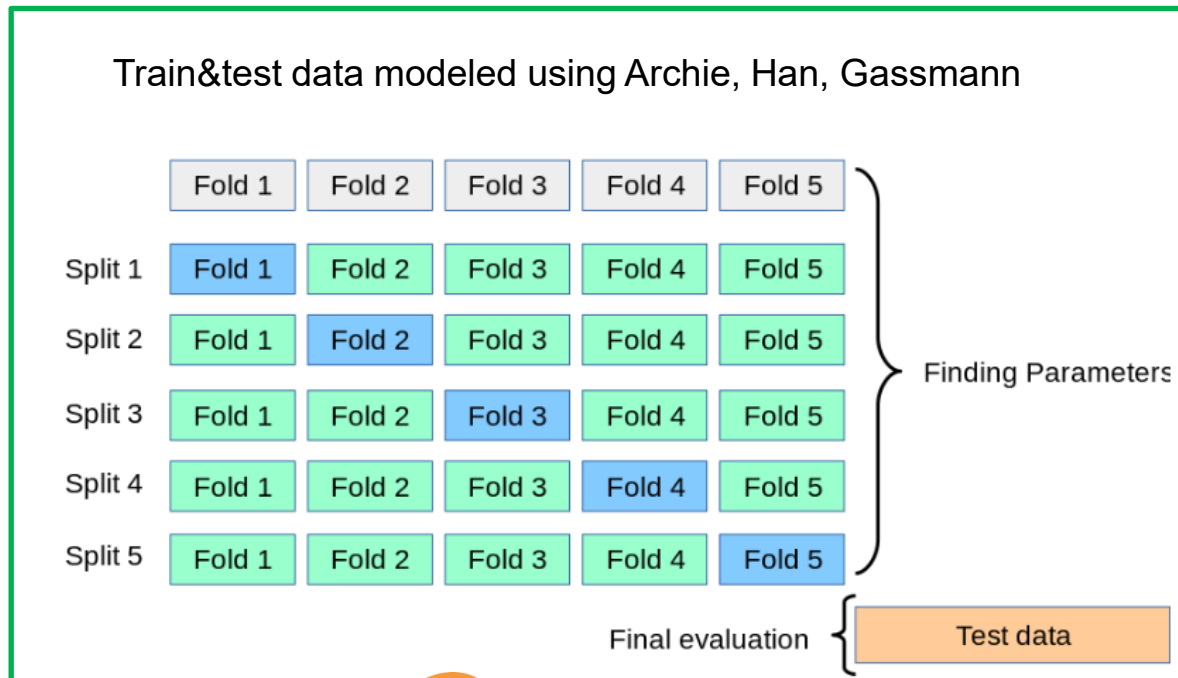
- Compute $\mathbf{m}$ by matrix inversion
- Minimize objective function w.r.t. $\mathbf{m}$
- $\mathbf{L}$ knows the physics

Machine learning



- Compute $\mathbf{L}$ by train and test.
- Minimize objective (cost, loss) function w.r.t. $\mathbf{L}$
- Labeled samples of $\mathbf{d}$ and $\mathbf{m}$

Solve the inverse problem using ML



1. Create labeled samples using rock-physics model
2. Train&test, 5-fold cross-validation
3. Predict on logs (or cubes)



Prediction on data



$\log R_T, v_p, rho$ logs (or cubes)

Computing uncertainty in ML predictions

Two-step methodology:

If model M_1 predicts target t from features $\mathbf{f}$ such that

$$\mu_{t|\mathbf{f}} \simeq \hat{t} = M_1(\mathbf{f})$$

then we can train a model M_2 to predict the variance as

$$\sigma_{t|\mathbf{f}}^2 \simeq \hat{s}^2 = (t - \hat{t})^2 = M_2(\mathbf{f})$$



Algorithm:

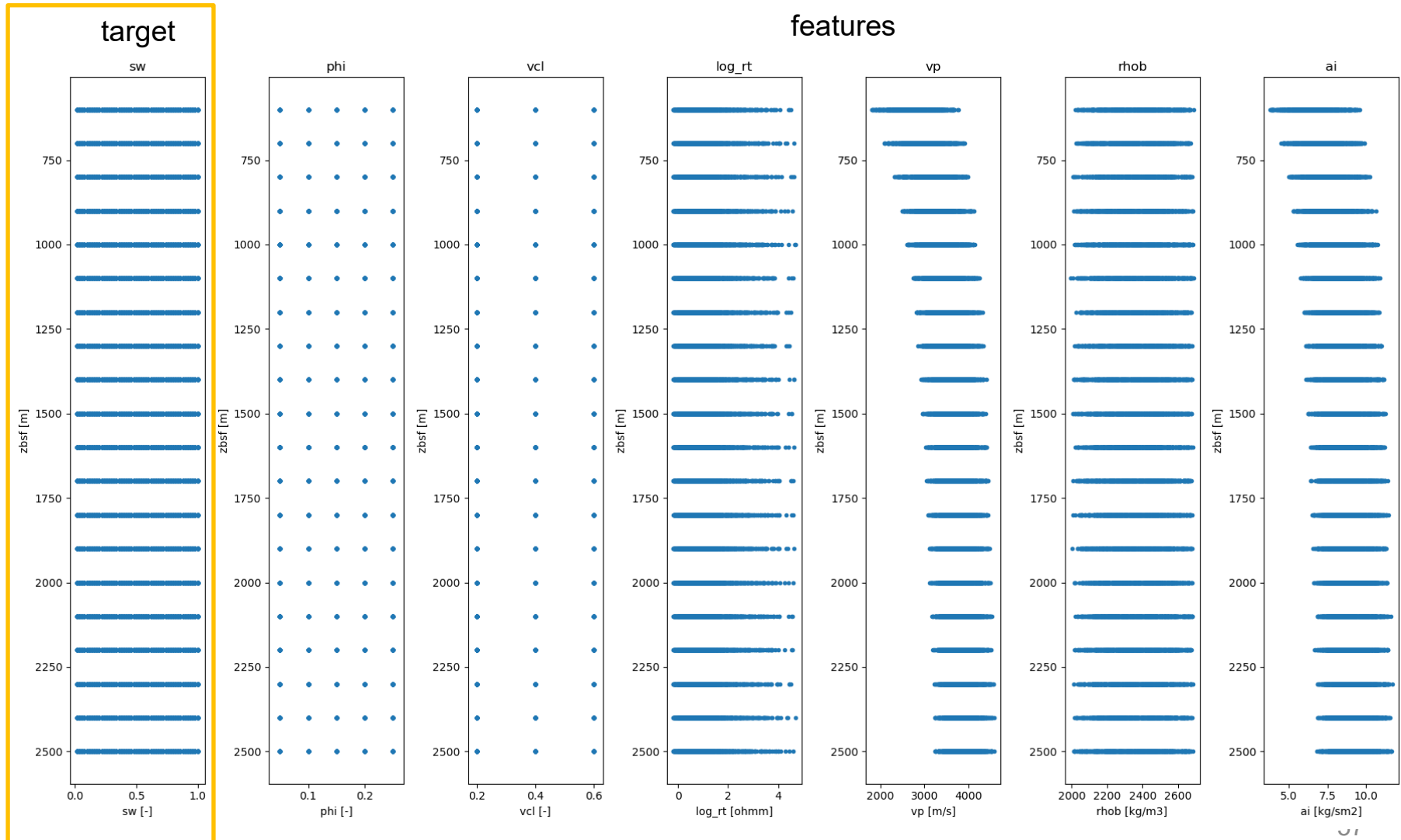
1. Split your train&test data in four subsets S_1, T_1 and S_2, T_2
2. Train model M_1 on S_1 to predict $\hat{t}_1$
3. Test M_1 on T_1
4. Compute prediction $\hat{t}_2$ on test set S_2 using model M_1
5. Train model M_2 on S_2 to predict $(t_2 - \hat{t}_2)^2$
6. Test M_2 on T_2
7. Apply your model to features $\mathbf{f}$, e.g.

$$t_{P10} = M_1(f) - 1.28\sqrt{M_2(f)}$$

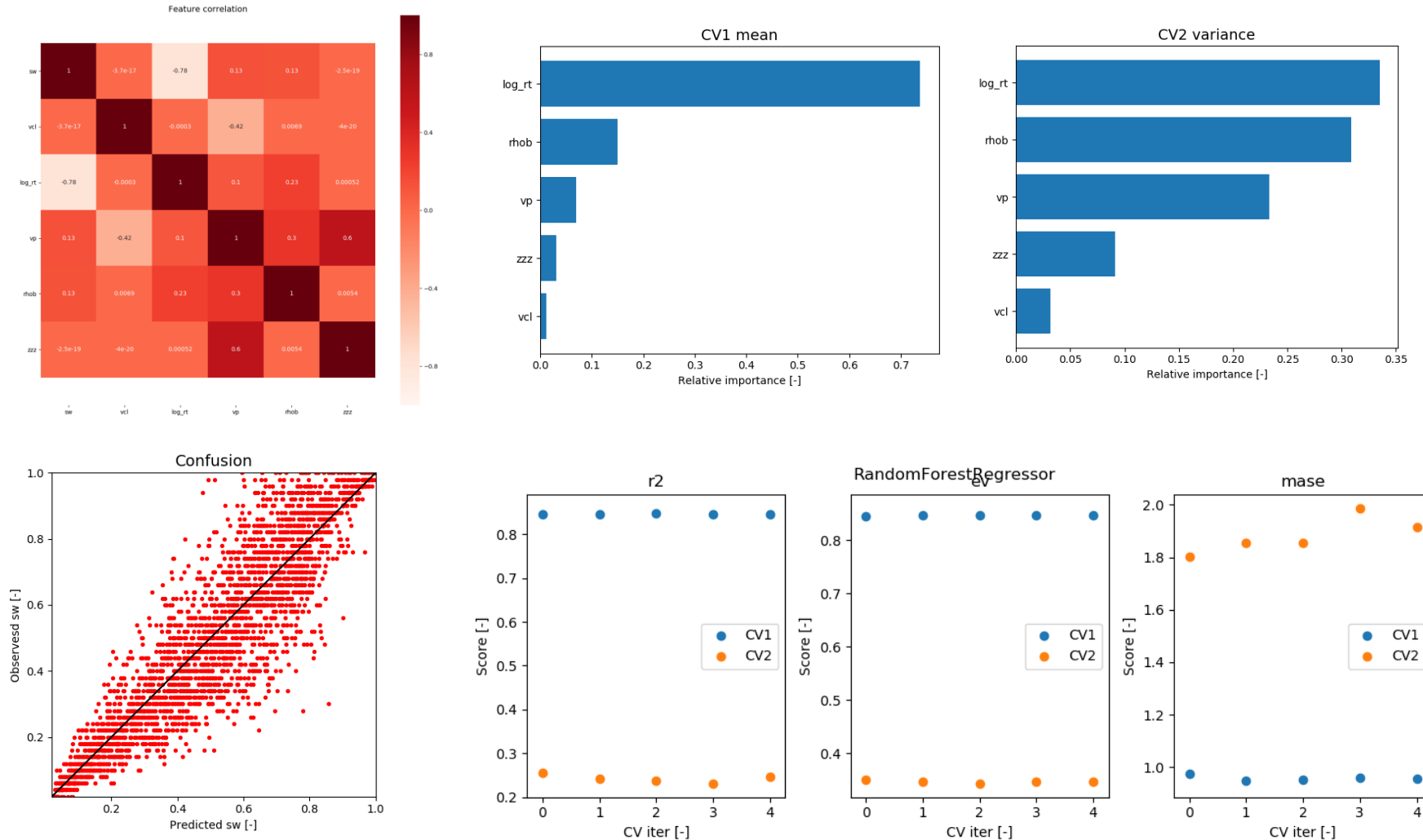
$$t_{P50} = M_1(f)$$

$$t_{P90} = M_1(f) + 1.28\sqrt{M_2(f)}$$

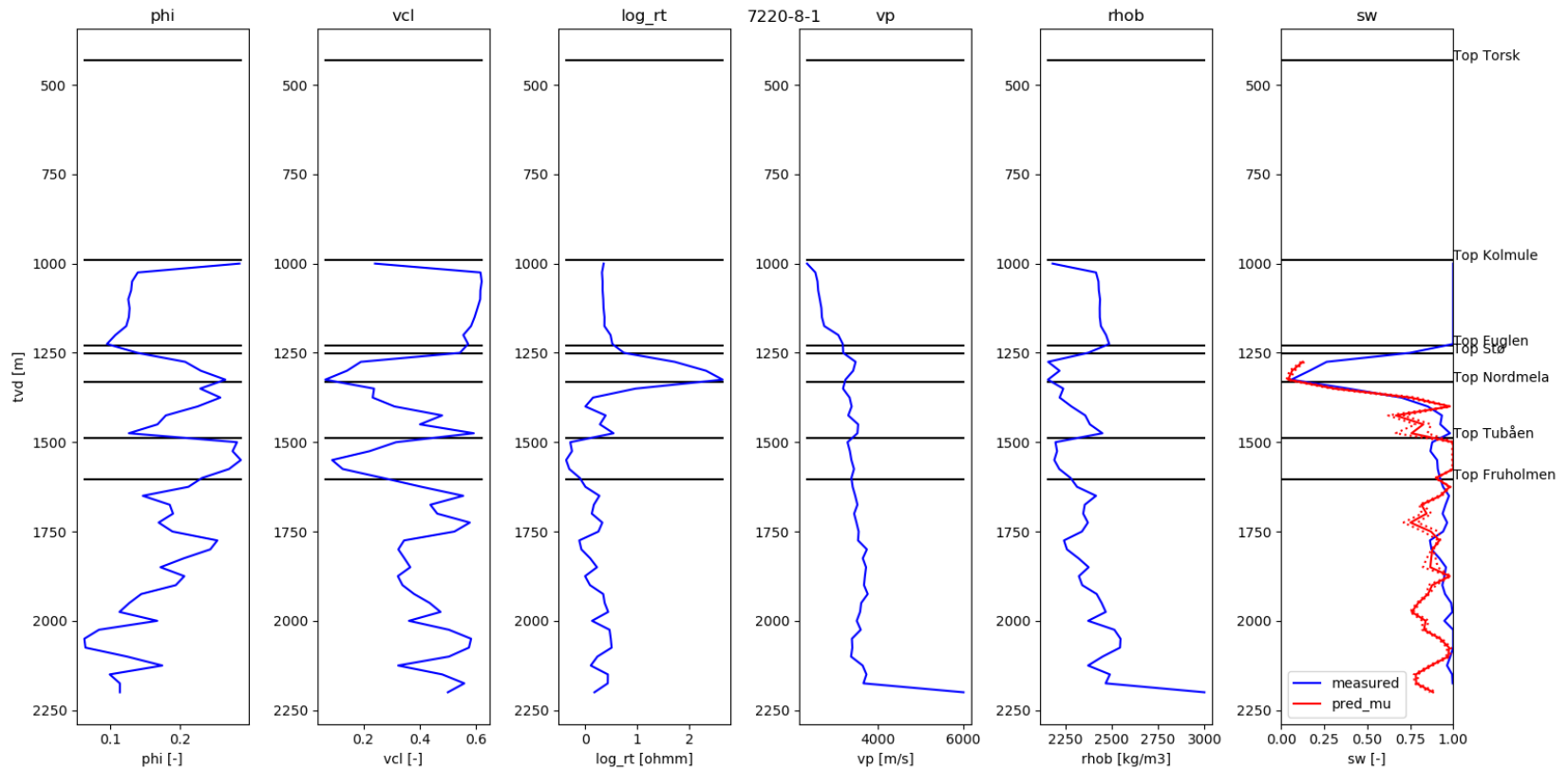
1. Craete label samples using the forward model (rock physics)



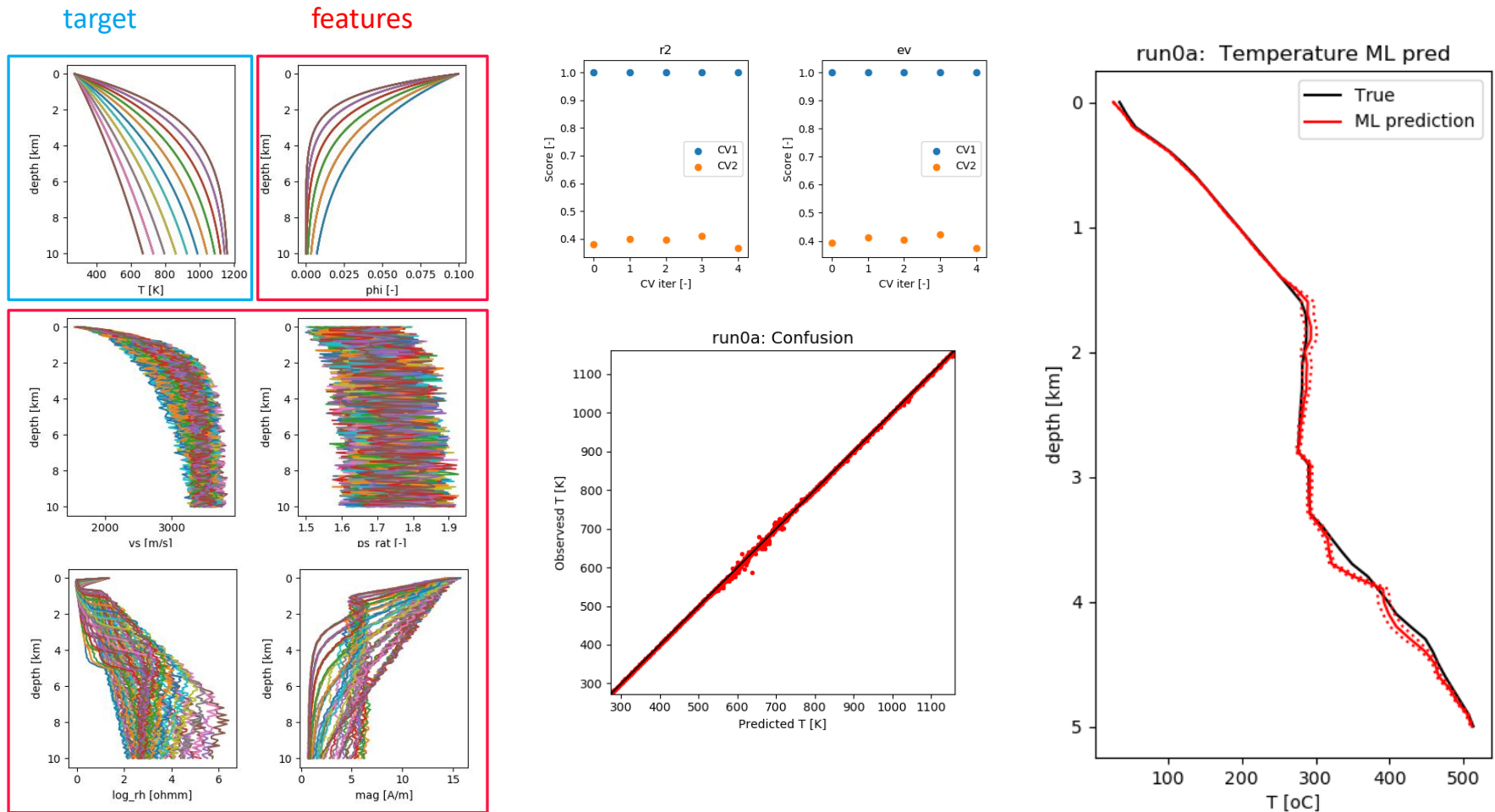
2. Train&test, 5-fold cross validation



3. Prediction on well logs



IDDP-2 synthetic test: Train&test data with noise



Random Forest prediction

Earthquakes, focal mechanisms

- Hypocenter: The location of the earthquake source (a fault)
- Focal mechanism: Character of the earthquake source

Seismic moment density tensor:

$$M_{jk} = c_{jklm}(x)n_l[u_m(x^+) - u_m(x^-)]$$

Strike slip: $\mathbf{M} = M_0 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Explosion: $\mathbf{M} = M_0 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Relationship with our previous work
on source-BC equivalence

$$\phi_j^M(x) = \int_{S_f} dS_f \partial_k \delta(x_f - x) M_{jk}(x_f)$$

S_f = fault plane

$$u_i(x) = \int_V d^3x' G_{ij}(x, x') \phi_j^M(x')$$

Earthquake source mechanism can be used to model piezo-electric transducers and viscoelastic waves in cores on centimeter scale

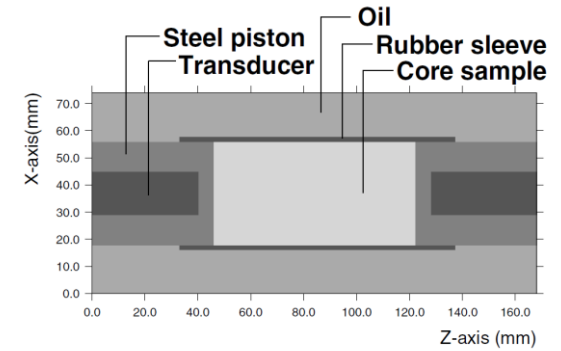


Fig. 1: Simulated experimental setup.

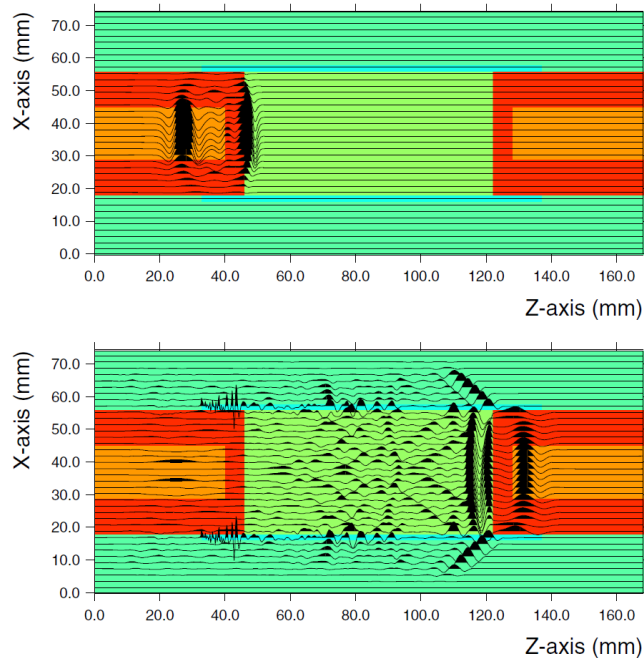


Fig. 3: Viscoelastic finite-difference snapshots for a P-wave source. Axial component of particle acceleration at time 8.0 and 36.0 μs .

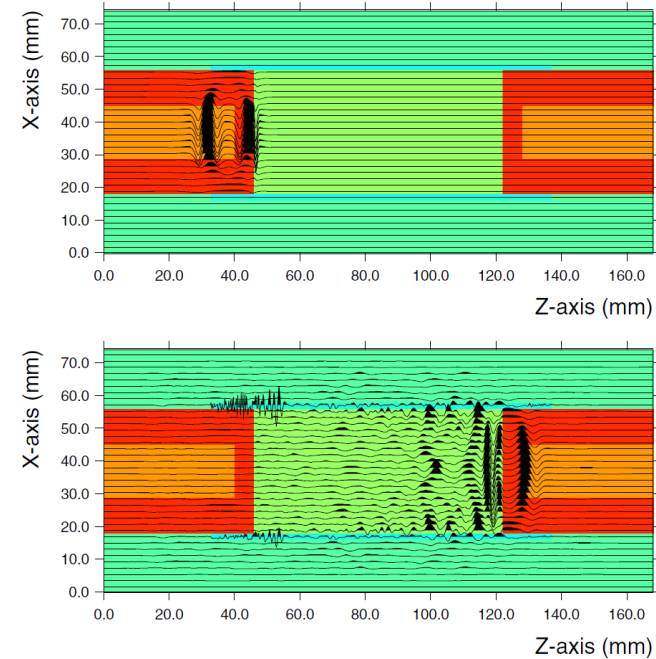


Fig. 4: Viscoelastic finite-difference snapshots for a S-wave source. Radial component of particle acceleration at time 8.0 and 58.0 μs .

Earthquakes, focal mechanisms

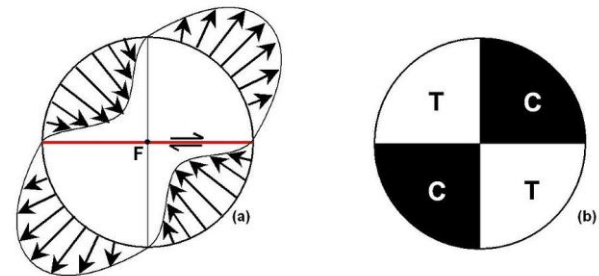
Strike, dip and rake angles

- Strike (ϕ): direction relative to north
- Dip (δ): angle from horizontal plane
- Rake (λ): Displacement direction relative to strike line (fault trace)

The moment density tensor can be expressed in terms of strike, dip and rake, e.g.

$$M_{xy} = M_0 \left(\sin \delta \cos 2\phi \cos \lambda + \frac{1}{2} \sin 2\delta \sin 2\phi \sin \lambda \right)$$

Beachballs

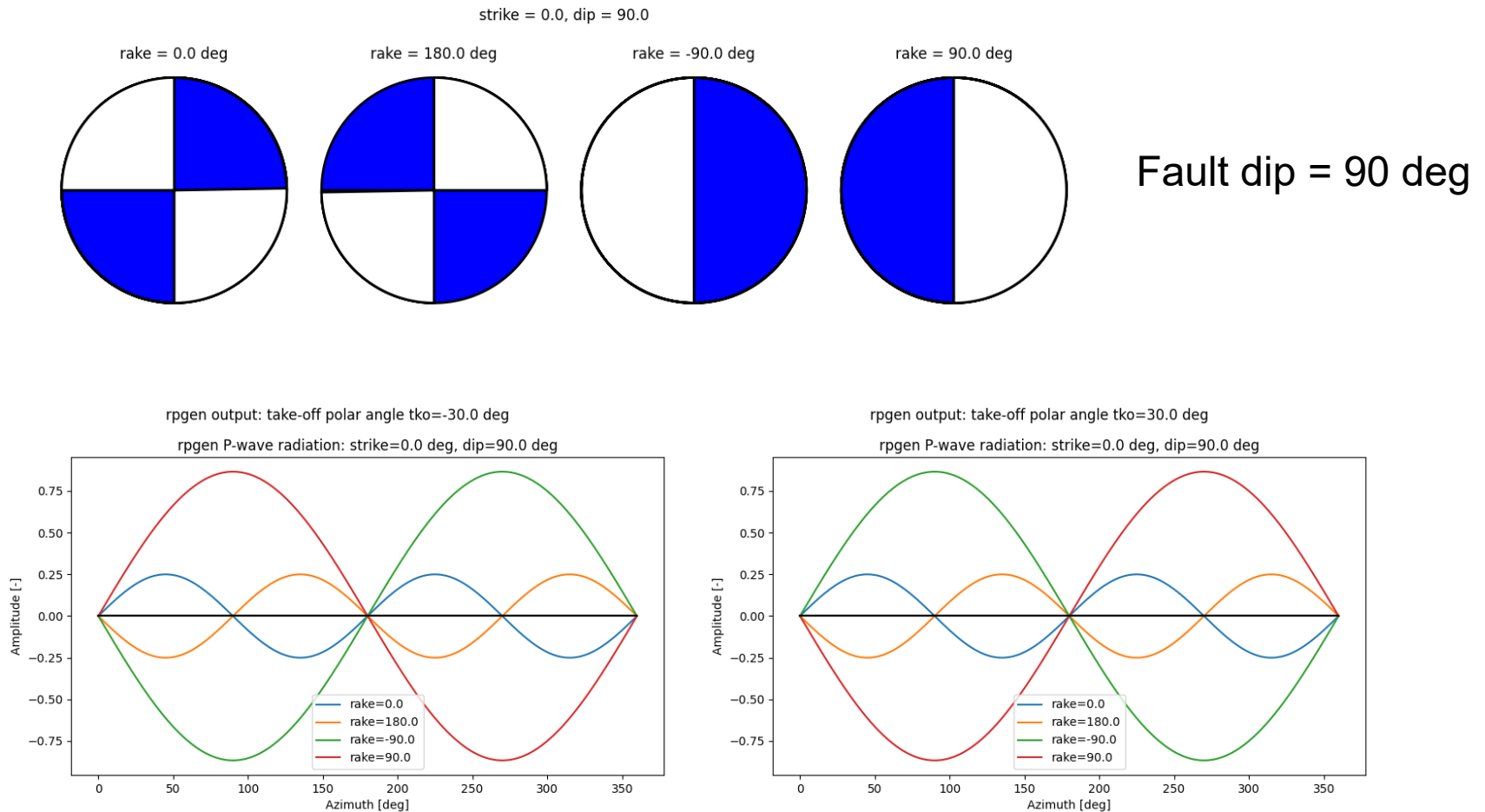


Fault types with corresponding beach ball plots^[2]

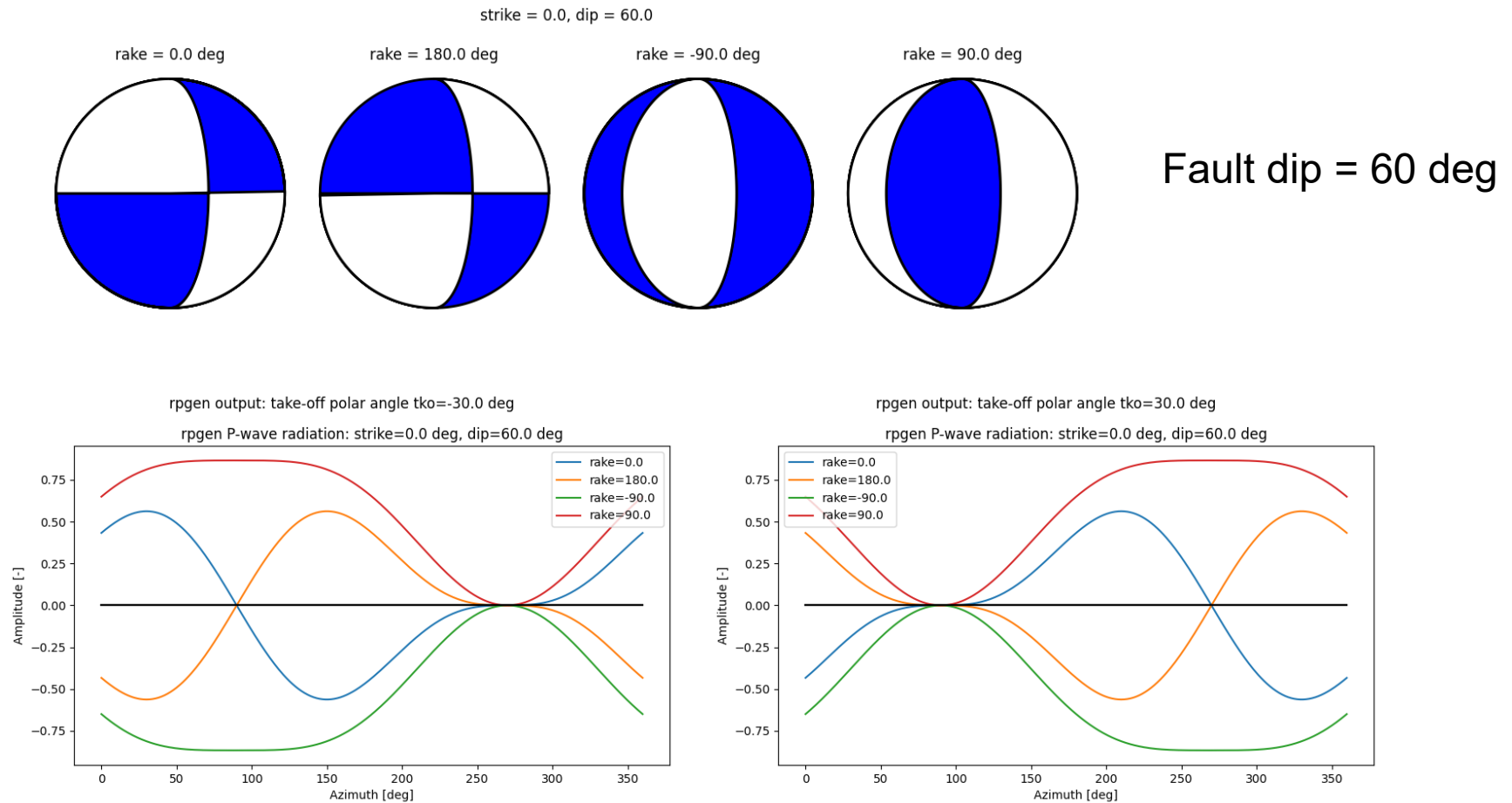
Left-lateral strike slip	Right-lateral strike slip	Normal dip-slip	Thrust/reverse dip-slip

Beachballs in Python:
`obspy.imaging.beachball`

P-wave first motion vs azimuth angle



P-wave first motion vs azimuth angle



rpgen function (Kwiatek and Ben-Zion, 2013)

Matlab ported to Python

First motion magnitude
depending on

- Fault strike and dip
- Rake angle (and tensile angle)
- Ray azimuth and take-off (grazing) angles
- P and S body waves

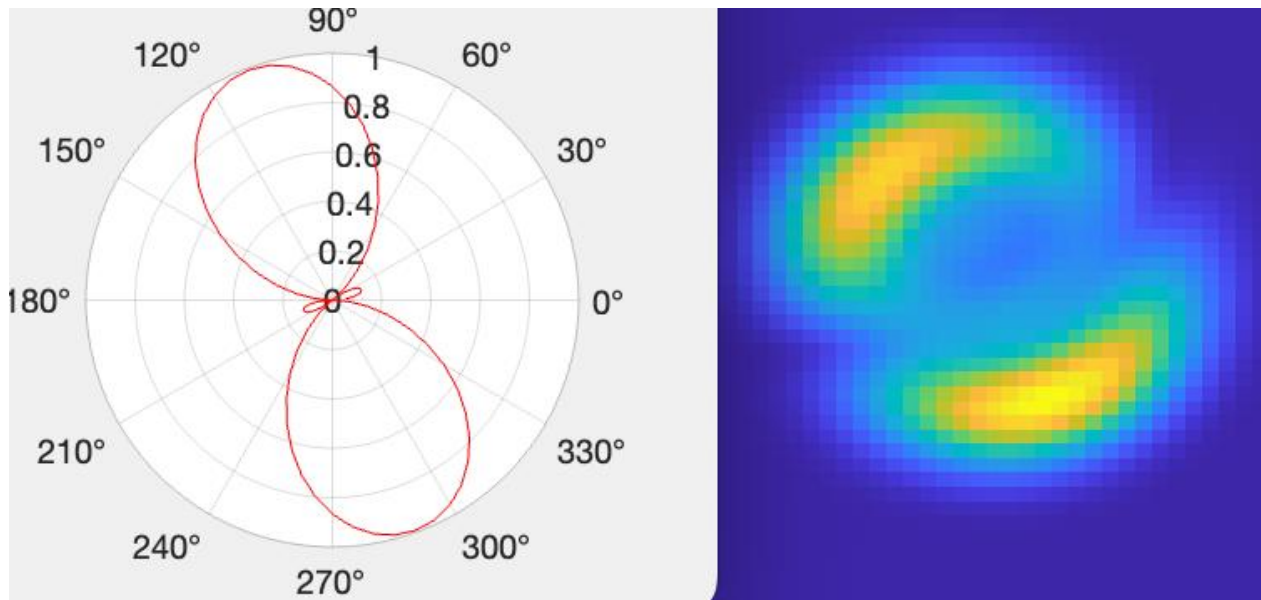
Reference:

Kwiatek, G. and Y. Ben-Zion (2013),
Assessment of P and S wave energy
radiated from very small shear-
tensile seismic events in a deep
South African mine. J. Geophys. Res.
118, 3630-3641

Kwiatek's Matlab code is public on
the Internet

```
Gp = cos(TKO)*(cos(TKO)*(sin(gamma)*(2*cos(dip)**2 - (2*sigma)/(2*sigma - 1)) + \
sin(2*dip)*cos(gamma)*sin(rake)) - \
cos(AZM)*sin(TKO)*(cos(gamma)*(cos(2*dip)*sin(rake)*sin(strike) + \
cos(dip)*cos(rake)*cos(strike)) - sin(2*dip)*sin(gamma)*sin(strike)) + \
sin(AZM)*sin(TKO)*(cos(gamma)*(cos(2*dip)*cos(strike)*sin(rake) - \
cos(dip)*cos(rake)*sin(strike)) - sin(2*dip)*cos(strike)*sin(gamma))) + \
sin(AZM)*sin(TKO)*(cos(TKO)*(cos(gamma)*(cos(2*dip)*cos(strike)*sin(rake) - \
cos(dip)*cos(rake)*sin(strike)) - sin(2*dip)*cos(strike)*sin(gamma)) + \
cos(AZM)*sin(TKO)*(cos(gamma)*(cos(2*strike)*cos(rake)*sin(dip) + \
(sin(2*dip)*sin(2*strike)*sin(rake))/2) - sin(2*strike)*sin(dip)**2*sin(gamma)) + \
sin(AZM)*sin(TKO)*(cos(gamma)*(sin(2*strike)*cos(rake)*sin(dip) - \
sin(2*dip)*cos(strike)**2*sin(rake)) - sin(gamma)*((2*sigma)/(2*sigma - 1) - \
2*cos(strike)**2*sin(dip)**2))) - \
cos(AZM)*sin(TKO)*(cos(TKO)*(cos(gamma)*(cos(2*dip)*sin(rake)*sin(strike) + \
cos(dip)*cos(rake)*cos(strike)) - sin(2*dip)*sin(gamma)*sin(strike)) - \
sin(AZM)*sin(TKO)*(cos(gamma)*(cos(2*strike)*cos(rake)*sin(dip) + \
(sin(2*dip)*sin(2*strike)*sin(rake))/2) - sin(2*strike)*sin(dip)**2*sin(gamma)) + \
cos(AZM)*sin(TKO)*(cos(gamma)*(sin(2*dip)*sin(rake)*sin(strike)**2 + \
sin(2*strike)*cos(rake)*sin(dip)) + sin(gamma)*((2*sigma)/(2*sigma - 1) - \
2*sin(dip)**2*sin(strike)**2)))
```

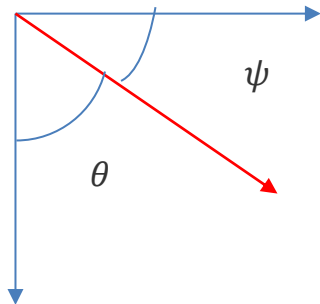
rpngen vs isotropic finite-difference modeling



Input parameters:
strike = 0;
dip = 45;
rake = 60;
sigma = 0.20; (Poisson's ratio)
TKO = 70;
AZM = 0 to 360;

Modeling by Ujjwal Shekhar, UiB

Before we can do inversion, we need to trace rays: P-waves in a 1D model



$$\frac{dx}{dz} = \frac{pv(z)}{\sqrt{1 - [pv(z)]^2}}$$

$$p = \frac{\sin \theta}{v(z)} = \frac{\cos \theta}{v(z)}$$

θ = polar angle

$\psi = 90^\circ - \theta$ = grazing angle

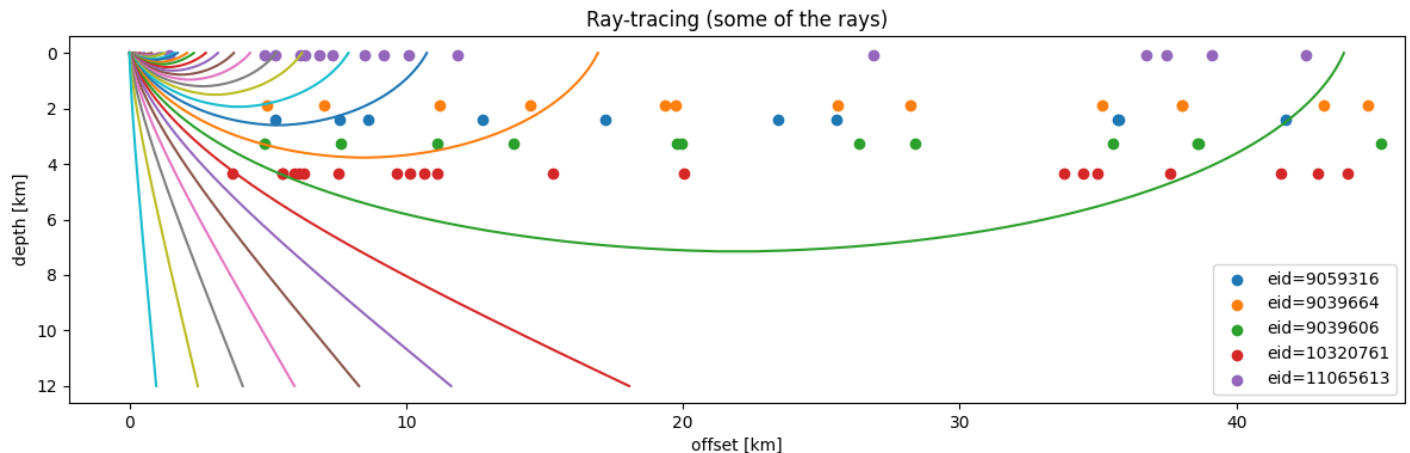
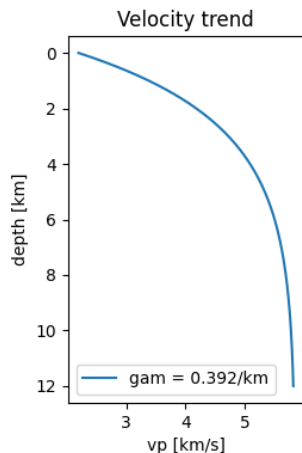
Luckily, there is Wolfram Mathematica:

Indefinite integral

Approximate form

St

$$\int \frac{p(a + (b - a)\exp(-gz))}{\sqrt{1 - p^2(a + (b - a)\exp(-gz))^2}} dz = \frac{1}{g} p \left(\frac{1}{\sqrt{1 - a^2 p^2}} a \left(\log \left(\sqrt{1 - a^2 p^2} \right. \right. \right. \\ \left. \left. \left. \sqrt{-a^2 p^2 + p^2(-(a - b)^2) e^{-2gz} + 2ap^2(a - b)e^{-gz} + 1 - a^2 p^2 + ap^2(a - b)e^{-gz} + 1} + gz \right) - \right. \right. \\ \left. \left. \frac{\sin^{-1}(p e^{-gz}(a(e^{gz} - 1) + b))}{p} \right) + \text{constant}$$



Rays (P-waves) are turning in the velocity gradient (ray-tracing using reciprocity)

Inversion of focal mechanisms

Bayesian inversion for strike, dip, rake

Posterior distribution

$$p(\phi, \delta, \lambda | d) \propto p(d | \phi, \delta, \lambda; \psi, \eta) p(\phi, \delta, \lambda)$$



posterior



likelihood

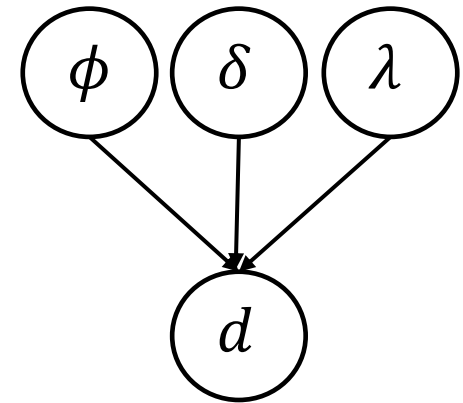


prior

Forward model (goes into the likelihood)

$$d = \text{sign}(G_P(d | \phi, \delta, \lambda; \psi, \eta)) + n$$

G_P is given by Kwiatek and Ben-Zion (2013)



Data:

$d = \pm 1$ (polarity of first motion)

Angles:

ϕ = strike

δ = dip

λ = rake

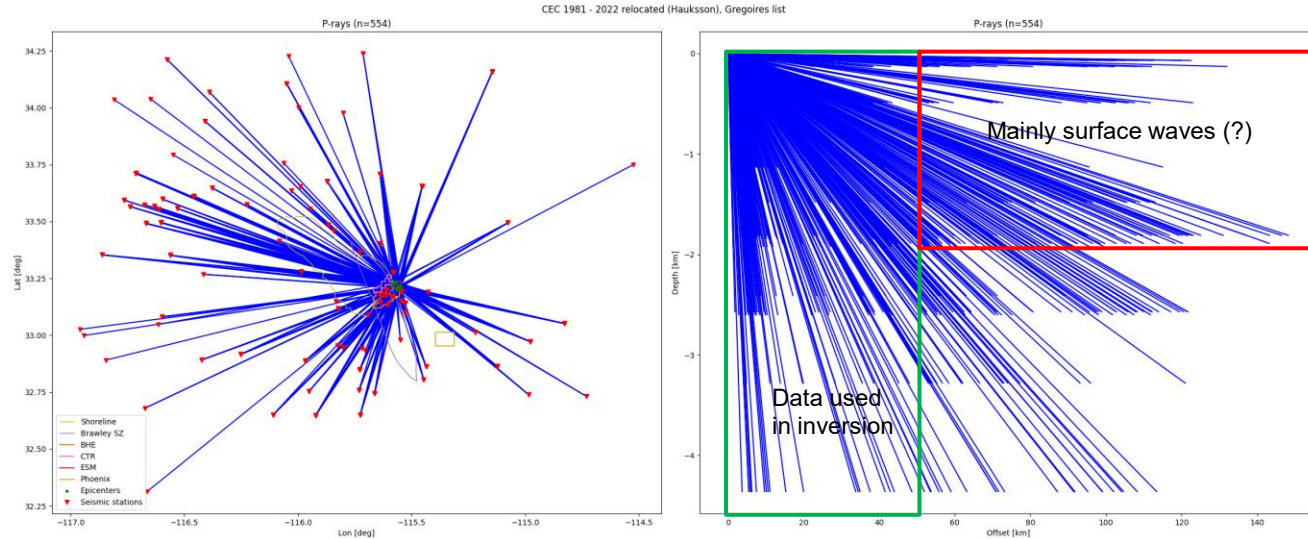
ψ = ray take-off
(grazing) angle

χ = ray azimuth angle

Data – selected earthquakes from SCEC (Hauksson, 2022)

Data from SCEC:

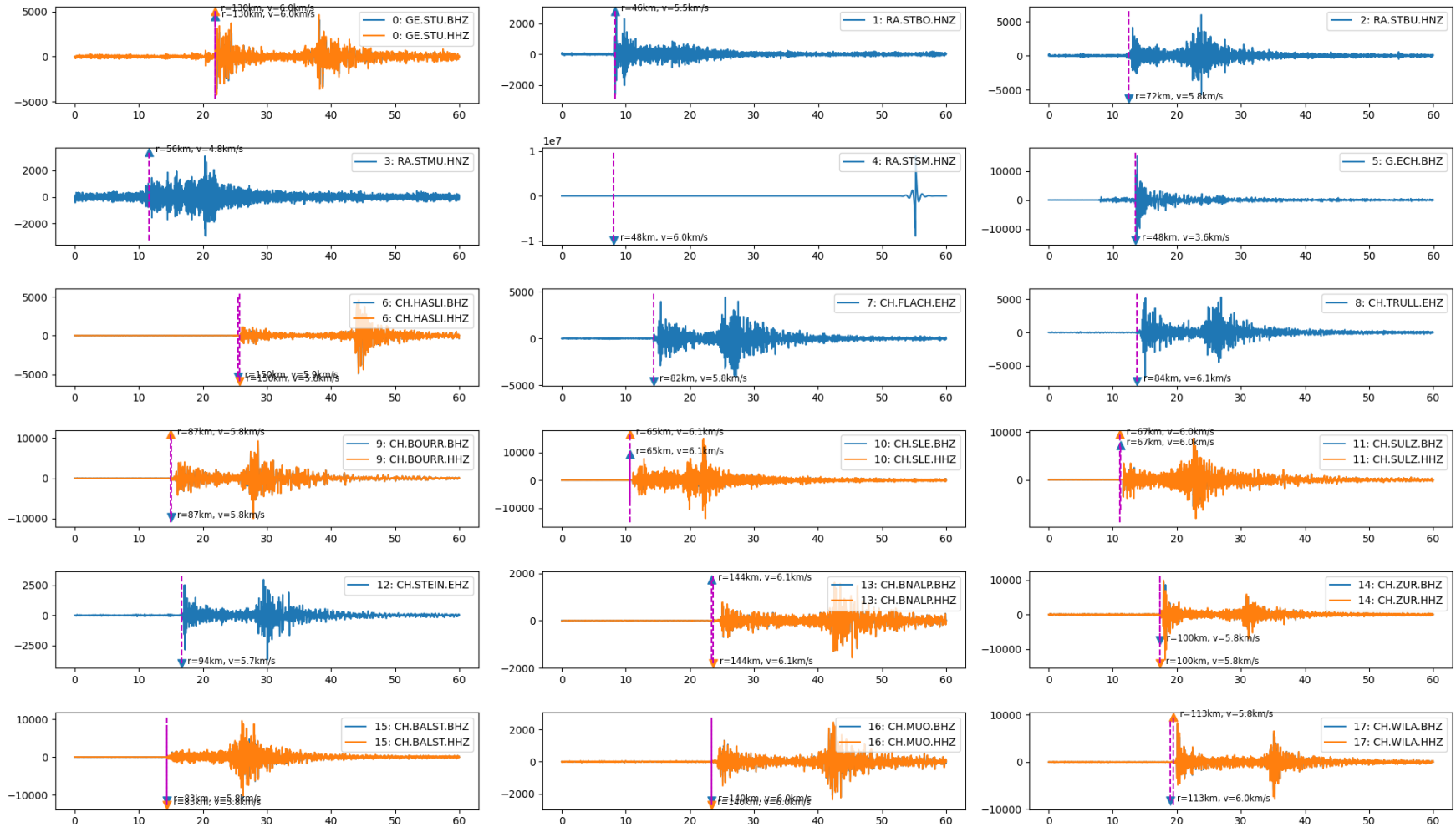
- Polarity of first P-wave motion (positive or negative)
- Max offset used in inversion: 50 km
- Events selected by Gregoire (geologist)



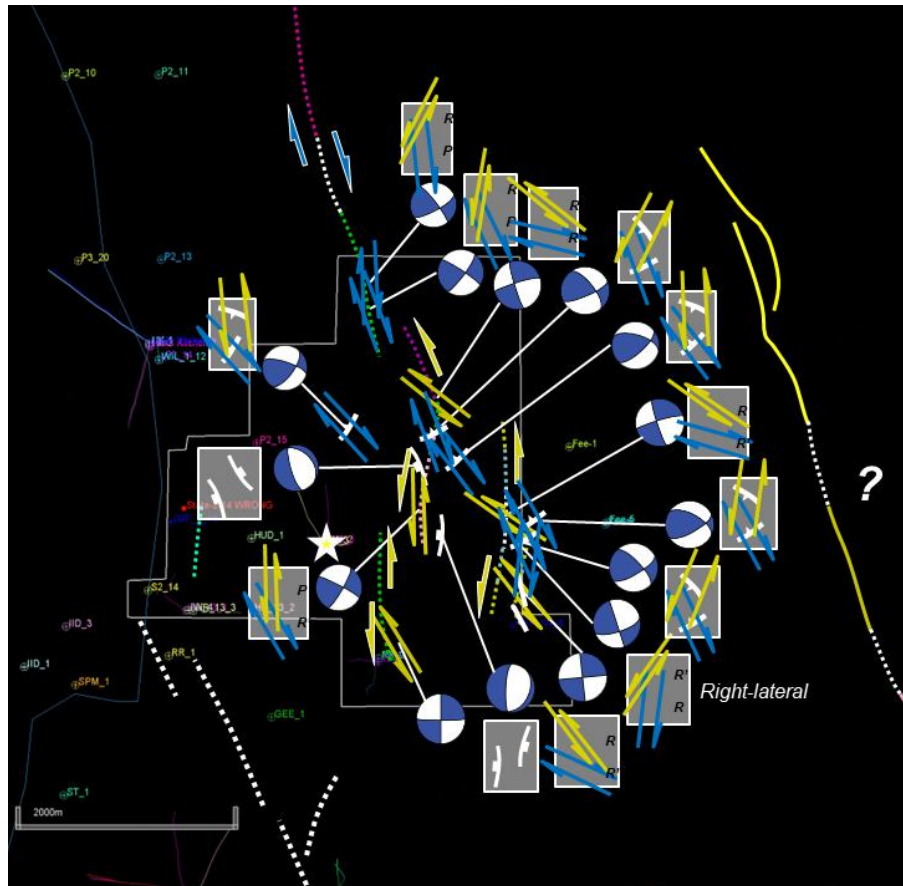
index	eid	etype	time0	lon0	lat0	depth0	mag	mtype	network	station	chan	lon	lat	elev	dist	phase	polarity	time	terr
0	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SSW	HHZ	-115.616	33.1775	-64	5.25	P	positive	1.179	0
1	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	CLI	EHZ	-115.527	33.1403	-92	7.58	P	negative	1.742	0
2	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	WI2	EHZ	-115.581	33.2768	-63	8.64	P	negative	1.882	0
3	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	IRS	EHZ	-115.429	33.1909	-43	12.77	P	positive	2.832	0
4	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	NW2	EHZ	-115.693	33.0873	-100	17.19	P	positive	3.782	0.01
5	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	FRK	EHZ	-115.638	33.4021	60	23.44	P	negative	4.162	0
6	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	ELR	EHZ	-115.833	33.1475	-97	25.53	P	positive	5.372	0
7	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SWS	HLZ	-115.8	32.9451	140	35.71	P	positive	6.382	0.01
8	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	SU2	EHZ	-115.8	32.9448	160.8	35.74	P	positive	6.432	0
10	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	COK	EHZ	-115.728	32.8495	-50	41.76	P	positive	8.032	0
11	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	CRR	EHZ	-115.969	32.8868	66	51.15	P	positive	8.972	0
13	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	BC3	EHZ	-115.453	33.6548	1080	51.62	P	negative	8.982	0
15	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	COA	EHZ	-115.123	32.8634	36	55.61	P	negative	10.132	0.01
16	9059316	earthquake	1998-07-03T19:0...	-115.566	33.2	2.41	2.83	Mc	CI	LTC	EHZ	-115.077	33.4938	275	55.97	P	negative	9.482	0

Earthquake seismograms and P-wave picks

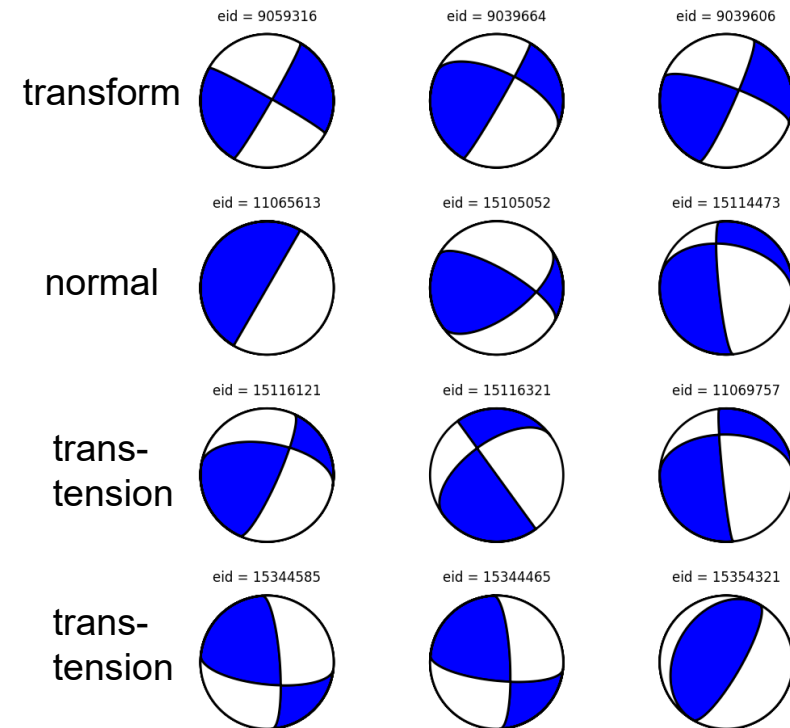
0: Event 223, 2005-03-14T12:34:59.964999Z, ML=4.0



Inversion result - beachballs

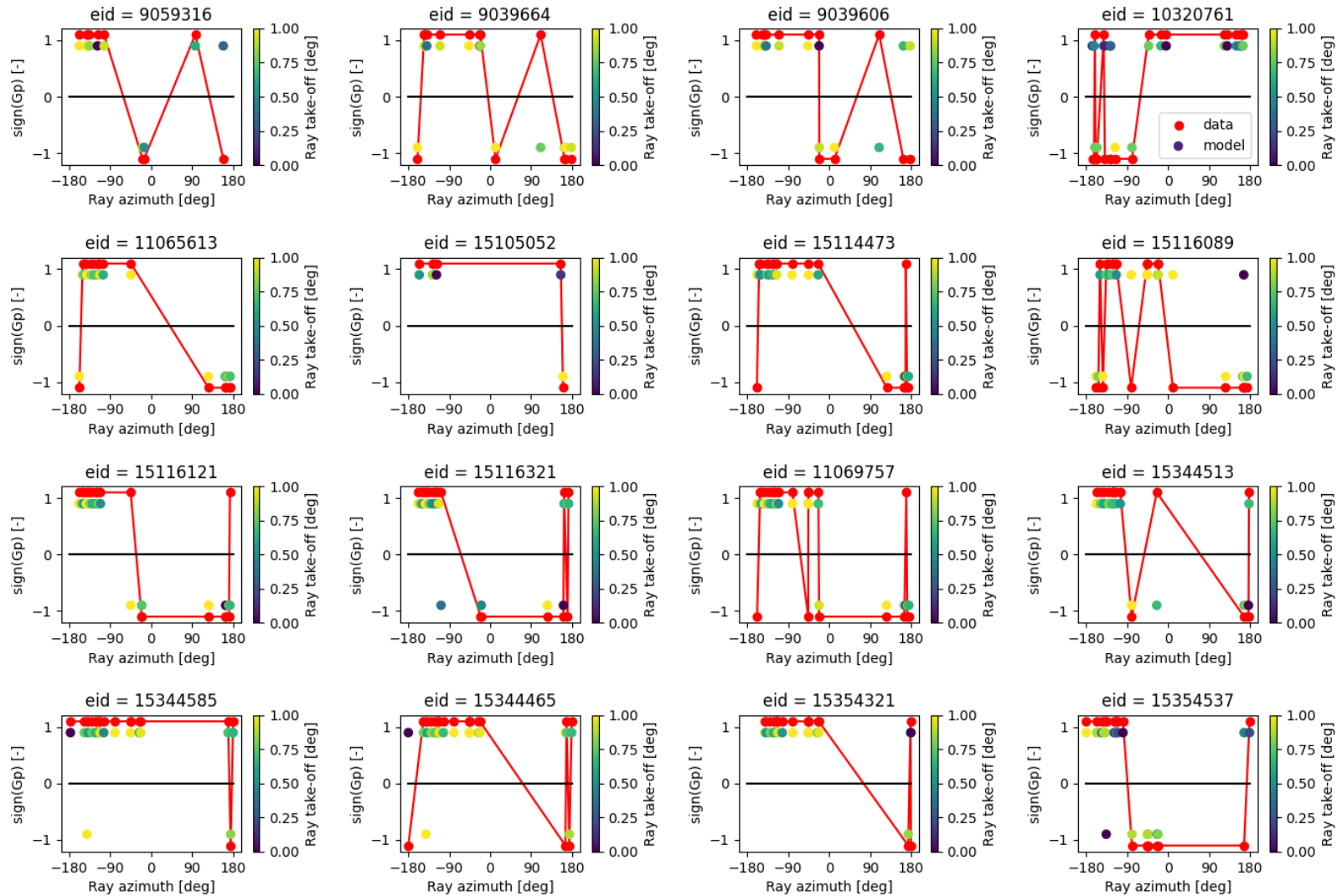


Focal mechanisms for selected events



Interpretation by geologist (Gregoire)

Data fit: Observed (red) vs forward modeled (colored) data



Example of Exam problem

Problem:

Write down the scalar Helmholtz equation, and explain the various terms. What is the difference for the seismic and EM cases?

Explain how one-way operators in the FK domain can be derived from the Helmholtz equation

Explain Claerbout's imaging principle. How does it relate to nonlinear inversion and the steepest decent method?

The solution to the Helmholtz equation can be written in terms of the Kirchhoff integral (representation theorem). Write down the Kirchhoff integral and explain the meaning of the various terms.

Explain how the Kirchhoff integral can be used to derive prestack Kirchhoff migration.

Mention other members of the Kirchhoff-family of migration methods, and some of their properties.